Acronyms Table

(except partners and advisory board members' short names found respectively in the frontpage and annex after section 5)

AB	Advisory Board
AWP	Agri-food waste plastics
BIC	Bio-based Industry Consortium
CAGR	Compound Annual Growth Rate
CAPEX	Capital Expenditure
CBR	Case Based Reasoning
CEP	Circular Economy Package
DD	Degree of Deacetylation
DSC	Differential Scanning Calorimetry
eBeam	Electron Beam
EIP	European Innovation Partnership
EoL	End of life
EVA	Ethylene vinyl alcohol
FCM	Food contact materials
FMC	Fortified microbial consortia
FP-insect	Insect fortified with PMC
FP-ew	Earthworm fortified with PMC
GA	General Assembly
GDP	Gross Domestic Product
GPC	Gel Permeation Chromatography
GHG	Greenhouse gas
HDPE	High Density Polyethylene
HSI	Hyperspectral imaging
IC	Innovation Committee
IDAE	Institute for Energy Diversification and Saving
IP(R)	Intellectual Property (Rights)
KPI	Key Performance Indicator
KER	Key Exploitable Result
KET	Key Enabling Technology
LCA	Life cycle *assessment*
e-LCA	Environmental Life Cycle Assessment
s-LCA	Social Life Cycle Assessment
LCC(A)	Life cycle cost (assessment)
LDPE	Low Density Poly Ethylene
LLDPE	Linear Low Density Poly Ethylene
Mtons	Million metric tons
MS	Milestone
MSW	Municipal solid wastes
NGO	Non-governmental organizations
NIR	Near-infrared
NMR	Nuclear Magnetic Resonance
NPK	Nitrogen, Phosphorus, Potassium
OPEX	Operational Expenditure
OTUs	Operational taxonomic units
PE	Polyethylene
PET	Polyethylene terephthalate
PLA	Polylactic acid
PM	Plastic Mulch
PMC	Plastic-degrading microbial/enzyme consortia
PNE	Plastic-degrading novel enzyme
PP	Polypropylene
PS	Polystyrene
PUR	Polyurethane
PVC	Polyvinylchloride
SEM	Scanning Electron Microscopy
SDG	Sustainable Development Goals
TGA	Thermal gravimetric analysis

1. Excellence

1.1 Objectives

Due to their unique properties, lightweight and versatility, plastics have outgrown most man-made materials. They are especially used throughout the agri-food value chain *(the waste thereof will be referred to in the proposal as AWP: agri-food waste plastics[1])*. From the globally produced 350 Mtons (million metric tons) of plastics per year, our target sectors consume in the range of 50% (packaging 40%, agriculture 3.5%)[2] and, in fact, they represent even a greater share of single and short-term use plastics. Nevertheless, the challenges linked to their end-of-life have long been under environmental scrutiny. Indeed, only 31% of plastic are currently recycled and an excessive 30-40% are landfilled, this being due to infra-developed technologies or to their unsatisfactory economic viability[3]. In addition, according to the Ellen MacArthur Foundation[4], as much as 32% end up out of controlled waste management systems and leak in our environment.

Plastic durability is a further challenge for agricultural films, which collection for materials recycling at end of life is broadly jeopardized due to their fouling. The affordable biodegradable plastic alternatives cannot degrade in the soil but could be sent to

1 Throughout the proposal we will refer to AWP for the plastics used in agriculture and food industries, those are for example mulching films or food packaging present in municipal solid wastes (MSW). Overall the project will develop biotech solutions for the degradation of 4-6 selected most representative fossil AWP and will be selected in WP1. In terms of MSW, we will focus on currently non recyclable mixed fractions such as plastic films, trays, etc.

2 https://www.plasticseurope.org/application/files/6315/4510/9658/Plastics_the_facts_2018_AF_web.pdf

3 https://ec.europa.eu/commission/news/first-ever-europe-wide-strategy-plastics-2018-jan-16_en

4 The new plastics economy rethinking the future of plastics, Ellen MacArthur foundation, 2016

composting plants adding a significant economic hurdle to their greater material cost and lower performance than mostly used fossil counterparts. As of 2017, approximately 6,300 Mtons of agricultural films plastic waste had been generated, around 9% of which had been recycled, 12% was incinerated, and 79% was accumulated in agri-fields (buried), landfills or the natural environment.
Likewise, more than half of the used food packaging collected in municipal wastes cannot be easily recycled due to the presence of leftover products and the fact that- as for certain mulching films and for most of food packaging, they contain complex laminated materials preventing their reprocessing. As such, most of the flexible and/or multilayer packaging are currently incinerated or even landfilled. This amounts for in the range of 13 Mtons per year in Europe[5].

RECOVER is a disruptive RIA proposal built upon the actual needs to solve the contamination of agro-fields with non-biodegradable agro-plastics from our end users as well as to improve municipal waste handling by decreasing drastically the packaging fraction going to unsustainable waste management routes.

We intend to reach our ambitious objectives applying biotech solutions that will include symbiotic bioprocesses between previously reported high powered microorganisms through microbiotic systems vectored by several insect species. Indeed, one of the main innovations in our proposal is the combination of endogenous and exogenous microorganisms and their inoculation on insects. This will allow achieving higher levels of biodegradation and direct conversion (at the same time) of AWP in insect by-products such as chitin-chitosan (high value as bioplastics raw materials and other interesting industrial applications).

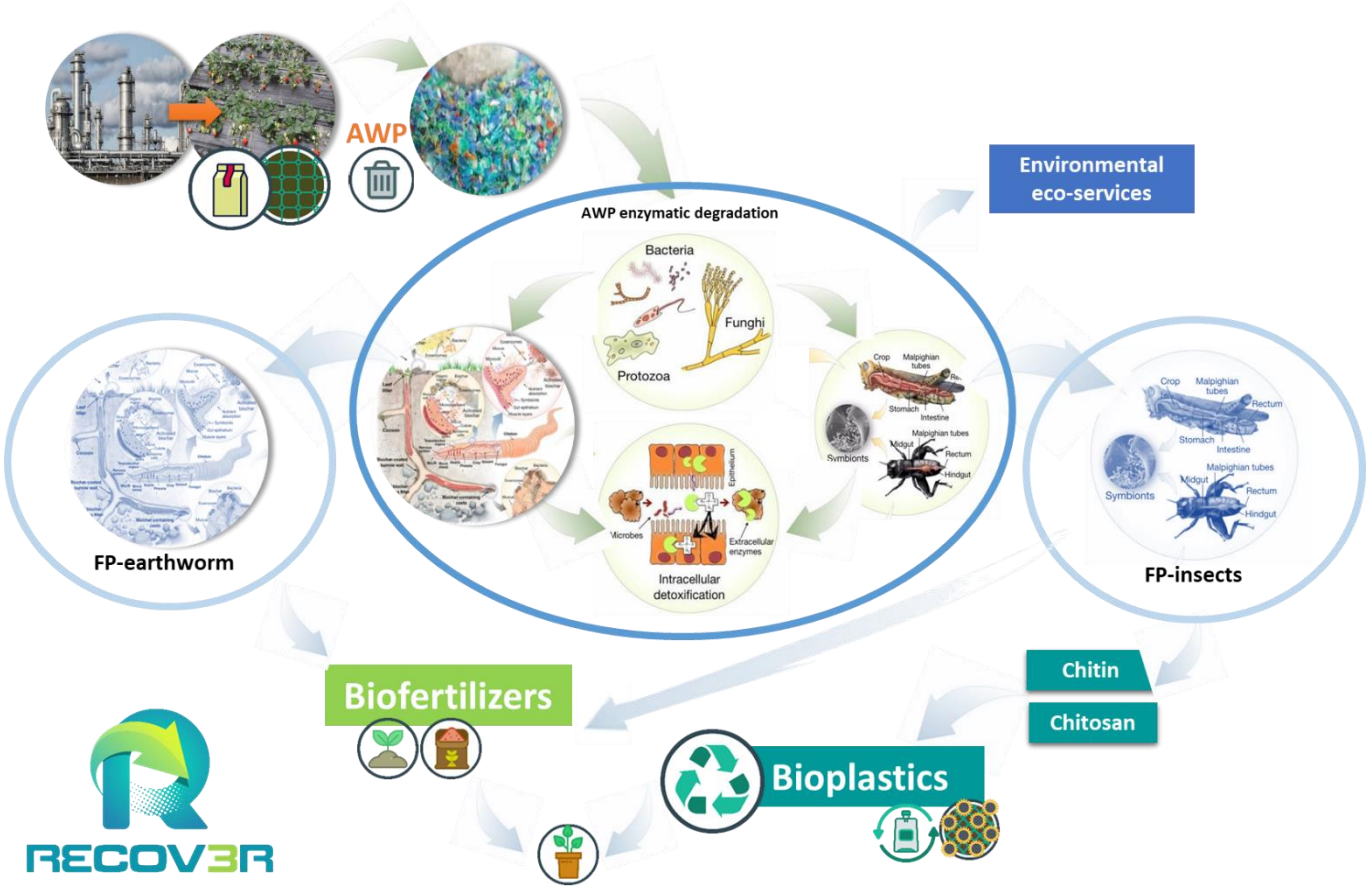

Fig 1.1: concept of the project and life cycle of our biotech tools

A great attention will be dedicated to establishing simple and reproducible environmental conditions that can be replicated in the farms or municipalities as a profitable activity that can benefit the whole value chains from plastic producers to the end-users involved in the project. As such, the business model, Life Cycle Assessment (LCA), Life Cycle Cost (LCC) and Technology Advanced Analysis (TEA) will be studied throughout the project. To validate the new metabolic routes and biotech tools, further certifications will be carried out regarding the actual biodegradation reached and innovative tools will be developed to support the implementation of the new process on suitable pre-sorted plastics fractions.
It is estimated that taping upon a share of the yearly used agricultural films and non-recyclable food packaging (43% of plastics

5 Gesellschaft für Verpackungsmarktforschung (GVM). Flexible Plastic Packaging Market in Germany and in Europe; GVM—Gesellschaft für Verpackungsmarktforschung: Mainz, Germany, 2017.

consumed in EU), the newly developed organic recycling route will help increasing plastic recycling in EU by 12 % (details in section 2) generating ~3.460 new jobs and an annual turnover of €724 million by 2030 (details specified in section 2). Furthermore, it will reduce CO_2 emissions derived from the AWP previously incinerated. The proposed bioremediation approach could be further expanded to recover plastic in existing landfills and highly polluted areas.

All in all, whereas there are huge challenges with the currently available end of life treatment of the plastics used in the agriculture and food industries, **RECOVER will help providing novel biotechnological solutions applying microorganisms, novel enzymes, earthworms and insects to degrade conventional plastic packaging and agricultural films waste streams but also result in new feedstocks for the bio-based industries.** The process will be optimized and performed either *ex-situ* in insect rearing chamber and composting reactors (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or *in-situ* in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment. In the former case, **the project will also deliver chitin/chitosan extracted from the vectorized insects after their use which can in turn be a new source of bioplastics used in our target sectors for e.g. improved mulching films and active packaging, as well as in new fertilizers valorising in cascade any by-products of the process**. In addition, RECOVER can solve the microplastics pollution at the level of both industrial composting and soil.

To achieve our goals a 48-month work plan with the following **specific objectives**, linked to their specific WPs and performance indicators (used as milestones for project implementation) has been defined:

Objective 1. To quantify and characterize AWP in order to select the proper ones for the project and define feasible logistic for collection, sorting and pre-treatment (Linked to WP1)

A comprehensive inventory and report on plastic types, collection strategies, cost, quality, safety and sustainability of AWP collection and possible pre-treatment will be made. All partners will be involved in this task and the collected data constitutes key elements to set the further approaches to treat AWP and RECOVER business strategies. ***KPI1:*** *by month 6 (MS1), a selection of 4-6 types of most representative fossil plastics (including in each case virgin plastics and AWP counterparts) picked individually from MSW or agricultural films will be available in order to provide normalized samples to all the WPs. By month 12 (MS2), obtaining the matrix for the definition of the classification, characterization and recommended biodegradation procedure. By month 12 (MS3), definition of the abiotic pre-treatment to be used depending on the mixture of AWP to biodegrade, and of micro-macro organism suitable for AWP biodegradation.*

Objective 2. To select microbial communities, novel synthetic enzymes, earthworms and insects for AWP biodegradation (Linked to WP2)

We will identify strains with greater capacity to use AWP as carbon source and explore the coding space of enzyme families in a focused fashion to combine independently evolved enzyme optimizations from different lineages of the same species, and from independent lineages that would have no capabilities to recombine and co-evolve enzymes in Nature. At least 4-6 species of insect and 2 species of earthworms for biodegradation of AWP will be tested. ***KPI2:*** *by month 12 (MS4), 3-4 microbial consortia and 2-3 enzymes selected. By month 18 (MS5), 2-3 microbial/enzyme pool for each earthworm and insect selected. By month 24 (MS6) novel enzymes and ne strategy for directed evolution available.*

Objective 3. To upscale the production of the suitable biotechnological solutions and combine them (Linked to WP3)

Stable plastic-degrading microbial/enzyme consortia (PMC) and worms and insects/larvae fortified with PMC for enhanced AWP degradation capability will be produced. ***KPI3:*** *by month 24 at least 2-4 stable microbial/enzyme inoculums (MS7) and 1-2 fortified insects and earthworms (MS8) will be available for the upscaling AWP biodegradation trials in WP4.*

Objective 4. To upscale and monitor biodegradation capacities of the fortified microbial consortia (FMC) (Linked to WP4)

The inoculum from the previous WP will be tested adding it in soil as bioremediation to AWP pollution as well as to feed earthworms/insects larvae permitting them to improve AWP biodegradation in bioreactors. Monitoring and modelling strategies will be put in place to optimise the *process.* ***KPI4****: by month 30 (MS9), process conditions for enhanced AWP biodegradation defined (MS9). By month 36 upscaled ex-situ and in situ AWP biodegradation processes performed (MS10) and biodegradation modelling established (MS11).*

Objective 5. To develop added value downstream products (Linked to WP5)

A green process will be set up to extract chitosan from the waste insects involved in the AWP degradation. The valorization of chitin and derived chitosan will be tested as additive in active plastic for packaging and mulching films providing anti-microbial and self-healing properties. Biofertilizer from vermicompost, insect manure and residues from chitosan extraction will be tested. ***KPI5:*** *by month 36 (MS12) 2-3 fertilizer products based on vermi-compost, insect and/or chitin available. By month 42 (MS13), extraction process for chitin from RECOVER by-products defined. By month 48 (MS14), 2-3 prototypes of products containing chitin (film, sheet, coated substrates) will be available and tested in food packaging and agriculture.*

Objective 6. To assess risks, safety, environmental impact, costs, logistic and cascade strategy for the RECOVER process (Linked to WP6)

Data collected during the project by all partner will be processed to perform analysis of risks and safety, in view of the implementation of the RECOVER AWP EoL. LCA, LCC and sLCA will be performed. The different RECOVER process scenario will be analysed to complete a techno-economic analysis, considering also the logistic aspects. Based on these results, the RECOVER cascade strategy for the full valorisation of RECOVER main and secondary products will be defined. ***KPI6**: by month 42 (MS15), a protocol for safe AWP processing and risk management and complete LCA results will be available to show the sustainability advantage vs. current EoL (target reduced CO_2 footprint by at least 50%)- **KPI7:** by month 48 (MS16), recommendations for RECOVER cascade process upscaling towards future full scale implementation available.*

Objective 7. To disseminate the project results and foster their efficient exploitation (linked to WP7)

In order to maximise its impact, dissemination and communication activities via traditional means (website, printed materials, press releases, etc.), scientific publications (peer-reviewed journals, conferences, etc.), specialised industrial workshops as well as via social media presence will be developed. An exploitation roadmap consolidating commercialisation plans and business opportunities, which will pave the way towards successful market entry of the products will be designed. ***KPI8:** by month 48 (MS17), the KPIs (e.g. organised workshops, participation to exhibitions and fairs, in-store boosts, newsletters, website, publications) listed in Section 2.2 (Table 2.3) will be achieved and a business plan including a roadmap for commercialisation of project results will be available. By month 48 (MS 18) achieve industry, stakeholders and policy guidelines.*

1.2 Relation to the BBI-JU Annual Work Plan

The Recover proposal, addressing the topic BBI-2019-SO2-R3, Apply microorganisms and/or enzymes to resolve end-of-life issues of plastics, will contribute significantly to the KPIs set out in the Strategic Innovation and Research Agenda (SIRA)[6] and is fully aligned with the BBI roadmap as will be further discussed in section 2. Our overriding goal aligns with most of its strategic orientations (SO) (fig. 1.2): SO1) The possibility to biodegrade plastic used in agriculture will make the crops growing more sustainable and significantly reduce the impact of land use and agriculture practices for biomass production; SO2) the core of our activities will deal with the setup of an efficient bio-technological (using enzymes, microorganisms optionally vectorised into insects) multistep process to convert fossil plastics into new valuable products; SO3) such as fertilizers, bioplastics and edible coatings materials which will be validated by relevant industry end users to reach SO4. Besides creating socio-economic value, the project will also help the EU in implementing better waste management strategies as well as reducing the major societal challenge reflected in the European Plastics strategy. We further cover the following cross cutting KETs: industrial biotechnology is at its core, advanced materials and manufacturing systems (converting waste into new bio-based feedstocks such as chitin/chitosan with support from advanced tools such as online monitoring and modeling). As part of our multidisciplinary approach, socio-economic science and humanities will also be considered where relevant, e.g. through the performing of a social-LCA, studying economics, sustainability as well as ethics aspects but also including marketing, policy making, training and upskilling in WP6-8 and also by getting public feedback from the dissemination and communication activities.

Fig 1.2: the 4 SO in the BBI SIRA 2017

The detailed relation of the proposal to the specific challenges and scope addressed by the topic are outlined in Table 1.1 below:

Table 1.1: How Recover relates to the scope and challenges of the BBI 2019.SO2.R3 topic

Topic text	How Recover addresses these challenges/scope
Include tailored microorganisms and/ or enzymes for co-digestion or co-fermentation of different feedstock, provided that the developed solutions result in sufficient yields in terms of: (i) breaking down the feedstock; and/ or (ii) conversion into valuable products.	Recover uses previously independently tested enzymes, microorganisms and insects that have shown the potential to degrade plastics (see details in section 1.4). The core proposal innovation lies on their novel combination aimed at maximizing the yield of the decomposition (insects) and degradation (microorganisms and enzymes). This will be achieved through the vectorisation of the later within the insects among other combined approaches to be tested. Earthworms are also incorporated in the compost as a synergistic agent to reach the ambitious proposal goal. The composting will be optimised possibly in presence of other biodegradable feedstocks (WP4). Furthermore, once our biodegradation agents will have finished their lifetime, the shell of the insects will be used to extract chitin, in turn converting a complementary fraction of the compost into new biobased materials available for further processing.

6 Strategic Innovation and Research Agenda (SIRA) BIO-BASED INDUSTRIES for Development & Growth in Europe – May 2017

To biodegrade/ decompose waste streams containing non-biodegradable polymers […] where standard EoL/ eco-design principles cannot be applied. May even be applied to all sorts of residual streams without any preliminary separation or sorting operations.	Recover focuses on fossil-based plastics used in agricultural and packaging sectors. Those mainly include PE, PET, PP, PS, EVA, etc. as well as laminates thereof among which at least 5 representative materials will be selected in WP1. Those are generally not bio-based, no intrinsically biodegradable and currently have a very low recycling rate (due to difficult recyclability in view of their fouling and /or complex structures). Our biotechnological degradation and valorisation process will be tuned to match the requirements of each of the selected materials employing or combining several approaches. We will first establish our process on separated plastics (sorted, and verifying the absence of contaminants that could affect the process) such as agricultural films which composition is generally known at the source as well as on selected food packaging. Based on the results, in WP6, we will also provide recommendations, how the case of comprising several materials can be approached (eg. combining several enzymes). Cascade processing will also be discussed for that purpose. On the long term, possibly performing further development for fractions not covered in this project, especially in case of actual MSW streams having very complex composition, it could be possible to apply a similar process without pre-sorting even of the MSW or for agri-waste of mixed organic and plastic types.
Apply the circular economy concept to plastics. Avoid plastic littering land and sea and reduce plastic waste diverted to landfill or incineration.	Finding value added EoL for plastics which are currently landfilled or incinerated (with a cost), or even far too often left on the soil, can act as the actual incentive towards the adoption of processes like Recover and significantly curb the current alarming figures. The potential revenues generated along the value chain, further discussed in section 2.2, support the way towards circular economy. This can help converting AWP to a feedstock with a value but also generate different products that can have their own markets. All is all, this will allow meeting the growing society's and users' demands to improve the EoL options to avoid landfilling and support manufacturers, who are responsible for the suitability of the materials they commercialise, to comply with relevant waste legislations. During the project implementation, benchmarking will be done vs. current EoL (often limited to landfilling or incineration) including economic as well as environmental sustainability studies of the delivered processes.
Remove non-biodegradable polymers from the environment.	Besides the plastics entering waste management systems that will be treated in novel industrial composting processes, our solutions will also be suitable for reducing the amount of non-biodegradable plastics reaching the environment in 2 ways: i) the *in situ* bioremediation of the agricultural plastics left on the soil will be extensively studied in field trials applying vermicompost based on a tuned combination of insects, enzymes and microorganisms (such an approach could also in the long term be adjusted as contingency for accidental littering in highly contaminated areas); ii) microplastics removal from compost, a critical problem for the current organic waste management companies, due to the presence of non-biodegradable plastic contaminants not properly sorted in the supposedly organic fraction collected from e.g. MSW.
Result in new feedstock for the bio-based industry.	In contrast we will convert non biobased AWP into a valuable feedstock for the BBI, and we will generate a number of semi-finished and finished value-added bio-enabled products to be commercialised across and beyond our value chain of actors such as tailored: -enzymes -microorganisms -insects & earthworms -vermicompost -fertilizers from by products of our process -chitin and chitosan from by products from our process with demonstrated potential to be used as new bioplastic packaging and to enhance mulching films and agriculture materials (pots, tips, etc). .
The TRL at the end of the project should be 4-5 for the bio-based value chain in question. Clearly state the starting and end TRLs of the key or technologies	The start, progress and end TRL of each technical component is justified in table 1.4. After characterizing the select AWP, we will work subsequently at lab (WP2) and pilot scale (WP3-5 covering the relevant steps along our value chain) obtaining significant amounts of biotechnological material for an industrially relevant validation of the new process. Different techno-economical aspects of scalability will be considered in WP6 to give recommendations for successful transfer of the different steps of our process to a sustainable industrial scale, so the results TRL is further matured to reach the market.
Industry participation in the project would be considered as an added value because it can play a supportive role to demonstrate the potential for integrating the developed concepts and technologies into current industrial landscapes or existing plants so they can be deployed more quickly and scaled up to apply industry-wide.	The whole circular bio-based value chain is covered by Recover partners. All WPs involve industrial partners, e.g. in the supply of AWP based on the initial industrial requirements defined in WP1, in the supply of enzymes and insects in WP2, in the upscaling of the biotechnological solutions for degrading plastics in WP3, in the scale up of both in situ and ex situ degradation trials, in the conversion/validation of the final value added product in WP5, as well as in the final sustainability and feasibility assessments (WP6). The consortium has a strong industry component including the participation of 7 SMEs, 3 Large enterprises and many more associated via membership in the participating industry association. It is noteworthy that 2 partners are associated BIC members and therefore committed to making the BBI roadmap successful. We cover the agricultural, food, plastics, fertilizer, biotech, etc. sectors. This dense ecosystem will allow researchers and early adopters alike to cooperate, understand the specific market demands, and set the basis for the long-term success of the new value chains. They will also drive the knowledge transfer across all involved sectors and the upscaling of the results obtained by the RTDs with a view to the smooth post-project replication. Our consortium, optimally constituted as a preliminary seed for the future value chain, involves a number

	of partners that are linked to end of life management and would be key in routing the core project results to the market: OWS RF (composting & biogas plant design), IDELUX (MSW management) and FEMTO (production of materials from recycled materials and RECOVER by-products). Furthermore, the solution can allow new actors such as farmers associated to ASAJA not yet involved in waste management to tap into the opportunity of converting their waste into value.
Proposals should address all requirements for RIA as shown in Table 3 in the Introduction of the Annual Work plan 2019	We comply with all requirements set in the AWP 2019: **Benefits versus the state of the art:** Those are discussed in section 1.4 showing our key progresses vs. the scientific literature available to date especially through TRL advances and vs. the industry status quo where AWP are not valorised. They will be reassessed as a conclusion of WP6. This activity will encompass the comparison of the Recover solutions with current EoL as the sustainability assessment. **Relationship with other projects:** Networking and clustering with other projects, clusters and initiatives will be addressed in WP7. A number of relevant ones are already discussed in section 1.3 and section 2. **Sustainability assessment:** WP6 includes an LCA following ISO standard ISO 14040 and ISO 14044, LCC and techno-economic analyses. Optimise resources even by studying best logistic aspects (reduce transport, energy consume etc.). Some initial environmental indicators regarding the current improper EoL approaches and replacement of fossil plastics with biobased ones are also discussed in section 2. **Compliance with relevant EU legislations & standardisation aspects:** Regarding the core biodegradation process for fossil plastics to be developed in the project, there are no pre-existing standards that would directly apply so wherever relevant we will back up on existing regulatory framework such as degradation standards ISO 14855 (compost) and ISO 17556 (soil). In order to positively influence the political framework for Recover exploitation, specific fact-based, scientifically backed information will be used in policy recommendations in WP7 to contribute to the discussion on emerging standards for biodegradation of such materials. To that end, we will provide a number of up scaled test results, to complement emerging scientific studies (see section 1.4). Regarding other aspects of the project, applicable standards will also be considered such as food contact requirements for the chitosan-based packaging, high-power raw materials for sustainable fertilizers, performance as pests control, etc. for agriculture applications. Although not mandatory requirements from the AWP 2019, as valuable to ensure a successful fate for our results towards the market, we will also consider: - Logistics aspects will be addressed in terms of sourcing in WP1 and in terms of different flows along our new value chain for cascade processing in WP5, and cascade strategies in WP6, based on the project results. - Business aspects are discussed including a preliminary outline of the investment needed for *ex situ* degradation solutions for farmers/agri-cooperatives as well as for MSW management organizations in section 2.2 The business plan will be much further explored during the project implementation in task 7.3 and 7.4.
Funding between 2 and 5 M€	In keeping with the expected range, the RECOVER proposal has a budget of ca €5.7 million and an EC contribution of ca. €4.4 million, therefore being within the range guided by the call text (including 23% in kind contributions by partners as described in section 2). The project will generate excellent value for money for the bio-based industries value chains while delivering a number of bio-based solutions.

1.3 Concept and methodology

Concept

Recover is a 4-year RIA project with 17 partners from 8 countries across Europe (Belgium, Italy, Spain, Germany, Portugal, Ireland, Luxembourg and UK). The consortium is very strong with plastic suppliers companies, innovative biotechnology and bioplastics companies and RTDs, as well as large end users. The proposal aims to develop biotech solutions for AWP treatment that will include plastic-degrading microorganisms, novel enzymes, insect larvae and earthworm which performance will be demonstrated in real plastic contaminated matrix both in *ex-situ* (composting, vermicomposting, reactor and insect rearing chambers) and *in-situ* (soil) experiments. In addition to compost and vermicompost a portfolio of added value products will be obtained including bioplastic formulations for application in agriculture (mulching films, sticks, pots), food packaging (trays, rigid containers, films) and coatings based on chitin/chitosan extracted from insects rearing on AWP; biofertilizers from organic matter leftovers after AWP treatments and chitin extraction. A great attention will be dedicated to establishing simple and reproducible conditions that can be replicated in the farms or municipalities as a profitable activity that can benefit the whole value chains from plastic producers to the end-users involved in the project.

While modern economies highly depend on plastics in Europe, in fact consumes in the range of 50 Mtons per year, only 25-31%

of the plastic waste in Europe is currently recycled and the larger share is incinerated or even landfilled[7] (high variability between countries). In the light of the EU's policy objective of achieving a resource efficient recycling society, it is hard to accept that in Europe we still landfill nearly 40-50% of plastic waste[8]. This all contributes to the depletion of limited resources and to the pollution of the environment. Besides those often non-optimal disposal routes, plastics, in particular packaging (bags, bottles, lids, etc.), account for 89-95% or roughly 8 Mtons of marine litter and 17 Mtons of plastic packaging is irreversibly dispersed on land every year worldwide[9]. Thus, the very slow rate of degradation of plastics leads to an accumulating pollution persisting for hundreds of years. Microbial biotechnology may contribute to this enormous challenge. The advances in our understanding of microbial functions from enzyme, to pathways, and entire metabolic networks now allow the engineering of complex metabolic functions in microbes[10]. With the ever-advancing tools from synthetic biology and the implementation of new strategies involving the use of microbial, insect and earthworm consortia, we can overcome the major challenges for plastic waste bioremediation and providing simultaneously a biotechnological plastic waste-based value chain for the synthesis of added value products, which will empower the recycling industry to a qualitatively new dimension. Proving biotech solutions for plastics EoL in our target sectors is of greatest relevance as they consume more than 40% of all plastic produced each year (packaging 40%, agriculture 4%)[11]. As such, more research and development has to be invested and more cost-competitive solutions have to be found to allow their eventual large-scale market-driven uptake.

The RECOVER project will:

- Develop a range of biotech tools with capabilities for degrading plastics wastes from agriculture and food packaging
- Validate the use of the resulting biotech solutions for the bioremediation of soil and organic matrices contaminated with plastic and for the biotransformation of plastics into added value products
- Progress in the use of the by products from the process (chitin, chitosan and organic leftovers) for the production of agricultural and packaging items and biofertilizers
- Assess at each process step, the environmental impacts and economic feasibility in comparison with existing benchmarks.
- Provide recommendations for suitable business approaches for the new process and resulting added value products from an integrated bio-based value chain perspective

The considered biotech tools are summarised in table 1.2 and the high added values applications in table 1.3.

Table 1.2: Recover biotech tools innovations to achieve project goals

Biotech tool	Effect/innovation	Environment of application		
		Compost	Reactor	Soil
Microbial consortia	Design efficient microbial communities with a balanced set of enzymes attacking mixed plastic waste under different environmental conditions	x	x	X
Novel enzymes	Development of better than nature enzymes that overcome the blockages for effective degradation of plastics that are highly resilient and unlikely to degrade naturally in the environment even those types currently recalcitrant to biologically driven breakdown	x	x	x
Earthworms+ microorganisms /enzymes	Explore and improve the natural capabilities for plastic biodegradation by earthworms	x		X
Insect larvae + microorganisms/ enzymes	Improve the natural capabilities of some insect known to feed on plastics and explore for new insect species capable to do so		X	

Table 1.3: RECOVER added value products and applications

Fraction	Product	Applications
Chitin from insect	Chitin-based bioplastics	Agriculture (pots, plant tips, etc)
		Food packaging
		Enhanced mulching films
Organic leftovers after chitin extraction	Biofertilizer	Agriculture
Vermicompost after plastic removal	Improved Vermicompost	Agriculture

7 http://ec.europa.eu/environment/waste/plastic_waste.htm
8 http://ec.europa.eu/environment/waste/plastic_waste.htm
9 THE NEW PLASTICS ECONOMY RETHINKING THE FUTURE OF PLASTICS, Ellen MacArthur foundation, 2016
10 Wierckx, N., Prieto, M. A., Pomposiello, P., de Lorenzo, V., O'Connor, K., & Blank, L. M. (2015). Plastic waste as a novel substrate for industrial biotechnology. Microbial biotechnology, 8(6), 900
11 Plastics the facts 2018, Plastics Europe, available at https://www.plasticseurope.org/application/files/6315/4510/9658/Plastics_the_facts_2018_AF_web.pdf

RECOVER will enable know-how transfer and business through its value chain of actors: from technology institutes & academia (5 RTDs: UAL, BRUNEL, UMH, UNIPI and ASU) to industry participants (7 SMEs), stakeholders and final users associations (ASAJA and SAV) and 3 Large enterprises leaders in their field (MANOR, FEMTO and SAV), biotech (ASA as well as CARBIOLICE and METGEN as industrial advisor), plastic converters (NATURPLAS, FEMTO), end users food packers (MANOR), organic recycler, technological solutions H&S and environment assessment providers (IRIS, OWS RF, IDELUX) and an innovation specialist (ENCO).
In RECOVER, our trans-disciplinary team combines all the relevant skills and organisations needed to address the full extent of the challenge of developing and validating innovative sustainable biotech solutions for processing plastic waste into added value products. The participation of a number of leading industry players directly and through the Advisory Board (AB) proves the actual soundness of the concept to generate large impact and will maximize the consortium capability in transferring this know-how to enable post project further demonstration and qualification into broadly available market ready solutions. Furthermore, the participating associations/NGO, covering the wastes management and farmers mainly, will also help raising environmental awareness and policy outreach that could further encourage the adoption of our new respectively *ex* and *in situ* solutions.
To complement our consortium, further stakeholders will be involved in an AB. METGEN and CARBIOLICE (provision of enzymes and their introduction in plastics), ISWA (international solid waste association), FCC (waste management), SOTRAFA (plastic producer and supplier), Biocampo (organic farming), CinG and CLT (fertilizers, compost & vermicompost), REPSOL (plastics), ALGAENERGY (CO2 capture), Remedia (scientific network for GHG monitoring), Packcluster (due to the high need of the packaging cluster to find complementary solutions for non-recyclable fractions) and COPA to disseminate the project results to the broadest farmers network in EU (support letters annexed to section 5), number of them (METGEN, SAV) from the BIC. They will be consulted as necessary for both technical, policy and business-related insights and further AB members may be appointed if needed.

Positioning of the project

Most activities performed in RECOVER project start at TRL3– experimental proof-of-concept or technology validation in lab (TRL4). To maximise our chances of success for the overall interdependent results, we have selected a number of complementary approaches as discussed in table 1.4.

Table 1.4: Summary of the TRL advancements that will bring about as result of the RECOVER project

Technology	Initial TRL	Final TRL	Comment
Selection of micro-organism and enzymes for plastic degradation, culture thereof	2	5	UAL have several projects in the area and the in silico studies of BRUNEL will allow fast selection of starting enzymes and MO
Vectorisation of insects and use for plastic degradation	3	5	NUTRINSECT had TRL9 for insect production so this knowledge will be coupled with existing lab results on insect degradation of plastics and their vectorisation with specific enzymes and microorganisms
Composting of conventionally non degradable plastics	4	5	OWS RF has a number of projects related with the field and also state of the art knowledge on biodegradation tests in different media that will allow taking existing results quickly to greater TRL
In situ bioremediation of plastic pollution	3	5	ASAJA, UMH and UAL have several points of testing for removing, pre-treatment and recycling and valorisation non-organic wastes. ASAJA is included in SIGFITO league for the promotion of plastic pollution removing in Spanish agri-fields.
Extraction of chitin assisted by eBeam	2	4	Aim is to apply eBeam irradiation to allow extraction by using more sustainable solvents. Only concepts available for now. Advances will be made in WP5 based on the results
Use of chitin and chitosan for food/ packaging and agriculture application	5	6	The objective will be adjusting results developed by UNIPI and IRIS, in use of chitosan and chitin in active packaging, to the new chitosan/chitin source and even to specific packaging and agriculture applications target of the project.
Validation of fertilizers from insects and chitosan extraction by products	2	5	Aim is the use of insect residues and chitosan/chitin in agriculture application, chitin/chitosan are known for action as pesticides
Cascade degradation and valorisation for optimal techno-economic feasibility	2	4	Only concepts are for now available. Advances will be made in the project WP6 based on the results and balances of each complementary approaches
Logistic system for plastic collection, separation and transport to treatment plants	2	5	The aim is to plan and test a logistic approach for collecting the plastic waste, separate the plastic types and direct them to the proper system of degradation

National or international research and innovation activities linked to the project

RECOVER will build on prior state-of-the-art as well as IP belonging to several project partners. A selection of relevant projects that the RECOVER partners are/have participated in are listed in Table 1.5, and derived expertise, methodologies, lab or pilot processes that will feed into RECOVER.

Table 1.5: Relevant research & innovation activities *(additional ones and details can be found in the respective partner profiles)*

Project	Partners involved	Opportunity for carryover to the project
AGRICHAR.-INNOVAI	UMH	Earthworms and biochar. Testing synergistic interactions to reduce pesticide impact in soils
ALTERNFEED	NUTRINSECT	Substitution of fish meals and oils by alternative products
ARBOCAR	ASA	Enzyme design and enzymatic modification of lignin for the refining of thermoplastic lignin compounds for car interiors
BIOCLEAN	OWS	New biotechnological approaches for biodegrading and promoting the environmental biotransformation of synthetic polymeric materials'', tasks OWS: biodegradability testing, bio-augmentation, bio-remediation
BIOnTop	ASU, IRIS, UNIPI, OWS, ENCO	Novel packaging films and textiles with tailored end of life and performance based on bio-based copolymers and coatings
COMPO-BALL	IRIS	Novel on-line composting monitoring system
DIBBIOPACK	UNIPI	Development of injection and blow extrusion moulded biodegradable and multifunctional packages by nanotechnology: improvement of structural and barrier properties, smart features and sustainability
FoodWasteTreatment	ASAJA	Integrated process for sustainable and cost effective food waste treatment
FORBIOPLAST	UAL, UNIPI, OWS	Forest resource sustainability through bio-based composite development
OPEN-BIO	OWS	Opening bio-based markets via standards, labelling and procurement'; task OWS: biodegradability in all kind of environments with specific attention to marine conditions, organic recycling (composting, anaerobic digestion), ecotoxicity
SYNPOL	OWS	Biopolymers from syngas fermentation'; tasks OWS: biodegradability and compostability (organic recycling) testing, ecotoxicity, LCA
RefuCoat	IRIS, MANOR	Full recyclable food package with enhanced gas barrier properties and new functionalities by the use of high-performance coatings
REGROW	UAL,UMH	"Reclamation of olive waste abandoned lagoons using bio-recovering strategies in a circular economy scenario", tasks UAL, UMH: Implementation of technology and biotech tools for the biodegradation of recalcitrant polluting compounds including the use of composting, vermicomposting, landfarming, phytoremediation and bioaugmentation with microbial inocula.
SABANA	UAL	"Sustainable Algae Biorefinery for Agriculture and Aquaculture". Tasks UAL: screening and development of microorganisms with biofertilizing, biostimulant and biopesticide effects
PlastiCircle	SAV	Rethink the different phases involved in transforming waste into valuable products: from the development of smart containers for separate waste collection to the improvement of transport routes and sorting technologies. The overall aim is to achieve recovery within the same value chain.
AGRIMAX	IRIS, UNIPI, UAL, FEMTO	biodegradable mulching films, fertilizers validation, biobased packaging, cooperative agricultural business models
AGROCYCLE	IRIS, OWS, MANOR, UNIPI	Production of biofertilizer, Valorisation of agrowastes
NANO-CHITOPACK	UNIPI (via INSTM)	Sustainable technologies for the production of biodegradable materials based on natural chitin-nanofibrils derived by waste of fish industry
POLYBIOSKIN	UNIPI (via INSTM), IRIS	Valorisation of chitin and chitosan for cosmetics and health care applications
MULTICYCLE	IRIS	Advanced & sustainable recycling processes and value chains for plastic-based multi-materials, circular plastics dissemination helix
ValorPlus	BRUNEL, ASA	Improving by-product output through sustainable and integrated closed loop biorefineries
VOLATILE	IDELUX, OWS	Biowaste derived volatile fatty acid platform for biopolymers, bioactive compounds and chemical building blocks
WaysTUP	SAV	Demonstrate the establishment of new value chains for urban biowaste utilisation to produce higher value purpose products (i.e. biobased products, included food and feed ingredients, biochar for application in agriculture) though a multi stakeholder approach in line circular economy

Overall approach and methodology

Our approach and methodology involve two main components:

1) **R&D and validation:** Research activities with the main focus to deliver Biotech tools with AWP biodegrading/biotrasforming capabilities within a number of scenarios. Besides, the development and the validation work will focus not only on the applicability of the solutions for AWP treatment and their scalability, but also on their environmental, social and economic sustainability using e-LCA, s-LCA and LCCA methodologies among others. The safety and compliance with other applicable standards and regulations of the developed treatment and products will be studied.
2) **Innovation:** This work will accompany and cross-cut the R&D work with the aim of preparing for post-project further development leading to the future replication of the project results. It consists of the following activities:
   a) **guidelines and recommendations for future full-scale implementation of the proposed technologies** via analysis of the **new value chains**, of the relevant ethical aspects as well as policy makers outreach.
   b) **dissemination** of the project results, including a series of activities to exchange knowledge between the stakeholders to validate their interest in the results of the project.
   c) **exploitation** and business planning to further increase the TRL of the developed processes and products beyond an expected final TRL5 to carry them over the notorious 'valley of death' towards commercial adoption.

How sex and/or gender analysis is taken into account in the project's content

The RECOVER consortium takes a comprehensive approach towards gender aspects. Gender is relevant to the content of the project because of the significant variability between the case of food packaging wastes and agricultural films ones. Indeed, women are the main actors in the sorting of household wastes but also have a prevalent role in decision-making in the food industry whereas the presence of females in the agricultural sector is traditionally limited, while there is a strong presence of females along the value chain of the bioeconomy industries. As such, innovation will consider the gender dimension and cater for the needs and motivations of females where possible. The dissemination and communication activities will therefore also be tailored to both genders to maximise RECOVER technologies acceptance. In order to foster the future of the circular Bioeconomy in Europe we will act for the education and training of the next generation of scientists, technologists and others jobs needed in the EU.

Besides in terms of gender dimension of the consortium, further discussed in section 3.3, the consortium is committed to follow the principles of gender mainstreaming in accordance with the Treaty of Amsterdam (1997) and other EU policy directives. It is noteworthy that there is a strong presence of female researchers, technologists, and industrials in the consortium, a number of which are occupying key positions in the project. A recruitment policy for future researchers and developers for the project is in place and is designed to ensure a reasonable level of balance between the sexes and afford them with equivalent opportunities.

1.4 Ambition

To support the European Strategy for Plastics in a Circular Economy[12] and contribute to alleviate the environmental problems faced by plastic waste contamination, RECOVER has the ambition to provide biotechnology-based tools (involving microorganisms, novel enzymes, earth-worm and insects) with plastic degrading capabilities that will be suitable for application in *ex-situ* (composting, vermicomposting, reactor, insect chambers) or *in-situ* (soil) treatment of plastic contaminated matrix. Biotech solutions will include symbiotic bioprocesses between previously reported high powered microorganisms through microbiotic systems vectored by several insect species. In addition, the outputs of these processes will be chitin from insects and biofertilizers from insects, earthworms, larvae etc. Our ground-breaking project objectives will lead to significant knowledge advancements to the state-of-the-art (SoA):

A. State of the art and advances in the field of plastic controlled waste management and treatment technologies

Plastic waste fractions pose a serious environmental threat due to its non-biodegradable and recalcitrant nature[13]. Besides, physical degradation of plastics in the environment produces secondary micro-plastics that cause even higher negative impact. Agroecosystems are likely to be the most plastic-contaminated terrestrial system outside of landfills and urban spaces. The source of these micro-plastics are recycled wastewater, biosolids and compost derived from non-source-separated residential waste or mixed municipal solid waste. Microplastics do not disappear naturally and, when mixed with soil, lead to decline of soil quality by hindering the transfer of water and fertilizer between soil and plants[14]. The final processing step for compost is typically sieving using 8-, 10-, or even 15-mm mesh sizes. Microplastics, defined as particles smaller than 5 mm, will therefore pass through these sieves and enter the compost. The quality and quantity of plastics to be recycled is a major obstacle to the development of this recycling process. The fact that the microplastics present in municipal waste consist of mixtures of different types of plastics, each

12 EU Strategy on Plastics: Protecting the Environment and Citizens from Plastic Waste (2018) https://www.interregeurope.eu/policylearning/news/3132/the-eu-strategy-on-plastics-protecting-the-environment-and-citizens-from-plastic-waste/
13 Moharir, R. V., & Kumar, S. (2019). Challenges associated with plastic waste disposal and allied microbial routes for its effective degradation: A comprehensive review. Journal of cleaner production. 208, 65-76
14 Ng et al. (2018). An overview of microplastic and nanoplastic pollution in agroecosystems. Science of the total environment, 627, 1377-1388.

consisting of a mixture of synthetic polymers, residual monomers and chemical additives adds a little more complexity to the problem[15].

The recycling of multilayer films, used broadly both in packaging and agriculture sectors, is still an important problem, which currently is not carried out in an industrial scale. There are several reasons for this, such as the large variety of materials used for each layer, large differences in the processing properties of the materials used for multilayer films, lack of systems for identification of multilayer film (e.g. PA/PE, PS/PE, PVC/PE, PET/PE, etc.), lack of system solutions for the collection of these materials, which results in storage with other foil materials with similar visible characteristics (colour, thickness, stiffness), lack of standard research of the properties, processing and applications of composites based on recycled multicomponent materials, lack of economically viable systems of segregation of the various materials. To some extent, the flexible packaging suffer similar difficulties in their recycling even when monolayer due to their difficult sorting and handling (separators are more suitable for rigid counterparts).[16]

The collection, transport, and treatment of these mixed plastic waste is difficult and often they end up in landfills, or in best option to incineration[17].

In RECOVER, with the help of plastic monitoring tools, equipment and logistic studies as stated in SoA below (C), a strategy for the targeted plastic waste collection and delivery to the appropriate degradation plant will be proposed and in the cases where a collection is not seen viable and an *in situ* (soil, compost) degradation is preferable, this will be highlighted (WP1).

B. State of the art and advances in the field of Biotech tools for plastic biodegradation

The biodegradation of plastics in the environment is a complex process that involves a combination of many abiotic and biotic factors. After fragmentation or modification accomplished by abiotic factors and microorganisms, the bulk polymer becomes more available for biological attack. Extracellular depolymerizing enzymes from microorganisms release monomers that are further intracellularly metabolized by them. Consequently, microorganisms and their enzymes are ultimately responsible for the complete mineralization, which is the transformation to carbon dioxide (CO_2), and the final removal. Research on the biodegradation of recalcitrant synthetic polymers with carbon-carbon backbones, namely PS, PVC, PP and PEs, has proved that they can be attacked, under different lab and actual site conditions, by a variety of mixed microbial communities and enzymes and pure fungal or bacterial strains obtained from terrestrial or aquatic environment and from some earth-worm and insects larval gut[18]. Earthworms and insect larvae are also able to degrade plastics in which process their gut microbiota play key role. Most work in this area has been focused on PUR and PET that are easier to biodegrade but biodegradation of other more recalcitrant plastics or mixtures are also scarcely covered[19].

Beyond the SoA, biotechnology-based solutions to mitigate the challenges associated to AWP management will be developed by RECOVER by providing biotech-tools (in WP2) that are able to biodegrade AWP, namely: Microbial consortia and novel enzymes with faster AWP biodegradation ability; Earthworms with improved capabilities for AWP biodegradation; Insect larvae with improved capabilities for AWP biodegradation. Such synergistic combinations were never reported before in the literature.

Bacteria reported for their prospective use in biodegradation of plastics include species of the genera *Pseudomonas, Streptomyces, Rhodococcus, Comamonas, Clostridium* and *Butyrivibrio*. Among the bacteria, one that stands out as potential candidate for plastic biodegradation is *Pseudomonas* because of its diverse metabolic and stress-resistance capabilities and genetic plasticity[20]. Several fungi also have the potential to degrade plastics in aquatic and soil environments. *Aspergillus, Fusarium, Pycnoporus* and *Mucor* are among those proposed for the biodegradation of plastics. In addition, fungi capable of degrading lignin in nature are of particular interest with respect to particularly inert structural elements of plastics such as C–C or C–Cl bonds. Lignin resembles certain plastics in not being hydrolysable and possessing similar structural elements that are oxidized and cleaved during lignin degradation by enzymes like laccase and lignin peroxidases that might enable them to degrade the C–C bond-based PE and PP[21].

To date, the rates achieved by pure microbial cultures have been low and almost all isolates are restricted to growth on a single type of plastic under optimal lab scale conditions[22] (Table 1.6). Consequently, **microorganisms and enzymes are needed that can more rapidly and efficiently degrade complex plastic mixtures**.

15Weithmann et al (2018)Organic fertilizer as a vehicle for the entry of microplastic into the environment, Science advances: Environmental Studies, 4(4), eaap8060.

16 Zenon Tartakowski, Recycling of packaging multilayer films: New materials for technical products, Resources, Conservation and Recycling, Volume 55, Issue 2, 2010, Pages 167-170, ISSN 0921-3449, https://doi.org/10.1016/j.resconrec.2010.09.004

17 https://www.amazon.it/Riutilizzo-materiali-polimerici-Mauro-Aglietto/dp/8868125242

18 Raddadi, N., & Fava, F. (2019). Biodegradation of oil-based plastics in the environment: Existing knowledge and needs of research and innovation. Science of The Total Environment 679, 148-158

19 Wierckx et al. (2018). Plastic biodegradation: Challenges and opportunities. Consequences of Microbial Interactions with Hydrocarbons, Oils, and Lipids: Biodegradation and Bioremediation, 1-29.

20 Wilkes, R. A., & Aristilde, L. (2017). Degradation and metabolism of synthetic plastics and associated products by Pseudomonas sp.: capabilities and challenges. Journal of Applied Microbiology, 123(3), 582-593.

21Krueger (2015). Prospects for microbiological solutions to environmental pollution with plastics. Applied Microbiology and Biotechnology, 99(21), 8857

22 Paço, A., Jacinto, J., da Costa, J. P., Santos, P. S., Vitorino, R., Duarte, A. C., & Rocha-Santos, T. (2019). Biotechnological tools for the effective management of plastics in the environment. Critical Reviews in Environmental Science and Technology, 49(5), 410-441

Table 1.6. Maximum biodegradation of synthetic plastics reported in literature for microbial pure culture[17]

Plastic	Microorganism	Weight loss (%)	Period
PUR	*Comamonas acidovorans*	48-100	7 d
PE/LDPE/HDPE	*Zalerium maritimum*	70	21 d
PET	*Ideonella sakiensis*	100	6 weeks
PP	*Pseudomonas stutzeri*	0.25	60 d
PVC	*Pleurotus sp., Phanerochaete chrysosporium, Poliporus versicolor*	0.55	30 d
PS	*Bacillus*	23	30 d
PC	*Phanerochaete chrysosporium*	5.4	12 m

Microbial consortia can improve degradation capabilities and promote the degradation of a more diverse group of plastics[23]. For example, the decomposition of micro-sized polyethylene (PE) by mesophilic mixed bacterial culture obtained from a municipal landfill sediment reduced the dry weight of particles (18% after 60 d)[24]. Also, microbial communities isolated from plastic garbage processing areas has been used for enhanced degradation of low density polyethylene (LDPE) strips and pellets with weight reductions of 81% and 38%, respectively, after 120 days[25].

The enzymes produced by microorganisms offer an additional alternative for plastic biodegradation. The enzymes reported to be involved in the breakage of polymers are cutinases, esterases, laccases, lipases and enzymes involved in lignin metabolism[26]. These are moderately active on hydrolysable polymers such as PET and PUR but no enzymes acting on the high molecular weight polymers such as PS, PVC are known[27]. There are a few examples of bacterial and fungal laccases and peroxidases that reduce the molecular mass of PE and cutinases and lipases with PET-degrading capacity. Using a combination of enzymes is also known to enhance polymer hydrolysis. Hydrolytic enzymes, are already produced in bulk quantities and low cost for a variety of applications. The French firm Carbios (to which Carbiolice, who in our AB and committed to collaborate with RECOVER, is affiliated) has optimized an enzyme that depolymerizes 97% of pretreated PET into monomers in 24 hours. The enzyme Carbios patent mentions bacterial and fungal cutinase and lipase enzymes[28]. Hence, the design of novel enzymatic degradation modules employing plastic hydrolases with high rates of plastic depolymerization is possible.

Beyond this, considering that the diversity of known enzymes and microbes acting on plastic polymers is still rather limited, their degrading activity is low and the action on complex plastic mixtures is not currently covered, RECOVER ambitions to explore the diversity of the cultivated and non-cultivated microorganisms (metagenomes) to seek for the building up of microbial communities and the identification of novel enzymes biocatalysts for AWP biodegradation. The search and final design of mixed tailored microbial/enzyme will be adjusted to each AWP degradation scenario, i.e, soil, compost, reactor or feeding earthworm & insect larvae.

As for the **design of novel enzymes**, knowledge-based design of enzymes is very poorly established, and the underlying understanding of structure-function relationships is so partial as to limit it substantially. Specific exceptions exist, such as strategies for P450 enzymes through biophysical property modification of active sites[29] (the basis of Oxford Biotrans technology) or modular construction[30]. Anyway these are rare and based upon modification of unusually well characterized enzymes. Frequently the largest influencers of enzyme activity do not lie within the active sites, and the most impactful random mutagenesis-generated improvements often lie in locations that could not have been predicted; and often are not understood even after they are identified. As a result, the principle source of new enzymes remains biological prospecting and screening for new activities and substrates.

We have identified metabolic enzymes with strong associations with higher fermentation productions and with increased substrate use (unpublished/commercially sensitive). This reveals that different evolutionary lineages are adapting with unlinked modular changes, and leads to a novel strategy for their recombination. The concept of enhanced recombination was established 20 years ago, when Pim[31] recombined four cephalosporinases and increased resistance 270-fold, and reviewed in[32]. This approach is

23 Drzyzga, O., & Prieto, A. (2019). Plastic waste management, a matter for the 'community'. Microbial biotechnology, 12(1), 66
24 Park, S. Y., & Kim, C. G. (2019). Biodegradation of micro-polyethylene particles by bacterial colonization of a mixed microbial consortium isolated from a landfill site. Chemosphere, 222, 527-533
25 Skariyachan, et al. (2017). Enhanced biodegradation of low and high-density polyethylene by novel bacterial consortia formulated from plastic-contaminated cow dung under thermophilic conditions. Environmental Science and Pollution Research, 24(9), 8443-8457
26 Narancic, T., & O'Connor, K. E. (2018). Plastic waste as a global challenge: are biodegradable plastics the answer to the plastic waste problem?. Microbiology, 165(2), 129-137
27 Wierckx, et al (2015). Plastic waste as a novel substrate for industrial biotechnology. Microbial biotechnology, 8(6), 900
28 Boisart, Cédric; Maille, Emmanuel. Method for recycling plastic products. U.S. Patent Application No 14/443,524, 15 Oct. 2015
29 Sowden, R.J. et al. (2018) Biotransformation of the sesquiterpene (+)-valencene by cytochrome P4509cam) and P450(BM-3). Organic and biomolecular chemistry. 3(1): 57-64.
30 Lapidoth. et al. (2005) Highly active enzymes by automated combinatorial backbone assembly and sequence design. Nature Communications 9: 2780.
31 Craeri, A. et al. (1998) DNA shuffling of a family of genes from diverse speces accelerates directed evolution. Nature 391: 288-291.
32 Carbone, M.N. & Arnold, F.H (2007). Engineering by homologous recombination, exploring sequence and function within a conserved fold. Current opinion in structural biology 17: 454-459

limited by the practicalities of molecular biology, species and sequence barriers to recombination, and identification of candidate variation for uses in new enzyme generation. We have developed tools and resources to identify the candidates affecting function, within and beyond the active sites (unpublished), that are part of tools originally intended to identify genes and polymorphisms that correlate with exploitable biological properties within bacteria.

Beyond the SoA, in RECOVER we will apply our unique comparative genomics tool to strains and species within and beyond our collections. This will be used to inform the design of synthetic sequences that optimally represent the behavior and activity determinants within enzyme classes. This combines novel genomics, with the latest synthetic DNA capabilities, and a new strategy for recombination that allows exploration of variations many orders of magnitude greater than was possible previously. A feature of this approach is that it permits multiple potentially cooperative changes to be generated in the new enzymes, traversing large parts of the enzyme fitness landscapes in single experimental steps. The goal is to both establish these new methods, which have wide potential applicability to green biotechnology and the sustainable bio-economy; and to generate new high-activity enzymes that target waste plastics as the specific objective of the project.

Earthworms provide multiple environmental services in agroecosystems[33]. These invertebrates exert a strong stimulating effect on soil microorganisms through their continuous ingestion of soil (endogeic earthworms) and leaf litter mixed with soil (anecic earthworms). In addition, the intense burrowing activity of these soil organisms contribute to disperse microorganisms in soil, so increasing the decomposing process of organic matter in the soil matrix[34]. Some studies suggest that these annelids may contribute to degradation of organic pollutants, thus resulting a complementary strategy of in-situ bioremediation[35]. Accumulating evidence suggest that earthworm interact with microplastics in soil[36]. Degradation of microplastics was demonstrated in the gastrointestinal tract of *L. terrestris* incubated in microplastic-contaminated soils. *L. terrestris* exposed to 7% microplastic LDPE (Low density polyethylene) could uptake LDPE and decrease its particle size thus facilitating the decay of LDPE as it passes through the gut. The bacteria consortia from the earthworm's gut were able to decrease particle size and metabolize it releasing long-chain alkanes as byproducts of bacterial decay[37]. However, these studies have been performed using soil-dwelling earthworms, and comparable interaction should be investigated in composting, epigeic earthworm species such as *Eisenia fetida*. Several studies performed with this genus have revealed that they build up their cast microbiome by selecting from the pool of ingested bacteria. A metagenomic analysis to characterize the bacterial communities of casts from *E. andrei* fed with different food sources found that the bacterial communities of cast strongly depended on the food source ingested by earthworms[38]. These promising results encourage the use of earthworms in the biodegradation of AWP wastes. This novel approach would increase the potential use of these organisms to manage plastic waste by farmers

Beyond this, RECOVER will explore the feasibility of inducing improved plastic degrading capabilities in soil-dwelling and epigenetic earthworms by strategies involving exposure to plastics contaminated environment and feeding or mixing with microorganisms. The improved earthworms are intended for mixed plastics biodegradation in soil and vermicomposting environments in WP4.

Larvae of some insects such as waxworms (*Plodia interpunctella)* and mealworms (*Tenebrio molitor*) are reported to feed on plastics and degrade them. In this process their gut bacteria play an important role in their ability to adapt to different foods and degrade plastics. Waxworms are capable of chewing and eating LLDPE films and two bacterial strains isolated from their guts, *Enterobacter asburiae* and *Bacillus* sp., were capable of degrading non-pretreated LLDPE (weight loss of up to 10% after 28 days)[39]. Mealworms readily ingest and biodegrade polystyrene (PS) to CO_2 and one bacteria isolated from their gut *Exiguobacterium* sp. was capable of PS degradation. This study showed that mealworms degrade nearly half of the ingested PS within the 12− 15 h in the gut, which is higher than the mass loss reported by the isolated *Exiguobacterium* sp. and other plastic-degrading bacterial isolates[40]. Recent work has also demonstrated that mealworm can also be fed on PE, PS and a mixture of both, and the microbiome present in the gut is necessary for the biodegradation; these two polymers were mineralized and/or incorporated into mealworm biomass. Analysis of the gut microbiome by next-generation sequencing revealed *Citrobacter* sp.

33 Blouin, M.; Hodson, M.E.; Delgado, E.A.; Baker, G.; Brussaard, L.; Butt, K.R.; Dai, J.; Dendooven, L.; Peres, G.; Tondoh, J.E. et al. A review of earthworm impact on soil function and ecosystem services. European Journal of Soil Science 2013, 64, 161-182
34 Yang, P., van Elsas, J.D. Mechanisms and ecological implications of the movement of bacteria in soil. Applied Soil Ecology 2018, 129, 112-120
35 Rodriguez-Campos, J.; Dendooven, L.; Alvarez-Bernal, D., Contreras-Ramos, S.M. Potential of earthworms to accelerate removal of organic contaminants from soil: a review. Appl Soil Ecol 2014, 79, 10-25
36 Huerta Lwanga, E.; Gertsen, H.; Gooren, H.; Peters, P.; Salánki, T.; van der Ploeg, M.; Besseling, E.; Koelmans, A.A., Geissen, V. Microplastics in the terrestrial ecosystem: Implications for Lumbricus terrestris (Oligochaeta, Lumbricidae). Environ Sci Technol 2016, 50, 2685-2691
37Puerta. Decay of low-density polyethylene by bacteria extracted from earthworm's guts: A potential for soil restoration. Sci Total Environ 2018, 624, 753-757
38 Aira, M., Olcina, J., Pérez-Losada, M., & Domínguez, J. (2016). Characterization of the bacterial communities of casts from Eisenia andrei fed with different substrates. Applied Soil Ecology, 98, 103-111
39 Yang, J.; Yang, Y.; Wu, W.-M.; Zhao, J.; Jiang, L. Evidence of polyethylene biodegradation by bacterial strains from the guts of plastic-eating waxworms. Environ. Sci. Technol. 2014, 48 (23), 13776−13784
40 Yang et al. (2015). Biodegradation and mineralization of polystyrene by plastic-eating mealworms: part 2. Role of gut microorganisms. Environmental science & technology, 49(20), 12087-12093

and *Kosakonia* sp. strongly associated with both PE and PS as well as species unique to each plastic. Consequently adaptability of the mealworm gut microbiome enables degradation of chemically dissimilar plastics[41].
According to these studies it is feasible to induce a specialized plastic-degrading microbial community in the gut of mealworms and to improve their capabilities by adjusting rearing conditions. Finally, the possibility of modifying and increasing capabilities of the microbiome within these organisms by feeding them with microorganisms is a subject to consider. Recently it has been shown that inoculating insects with symbiotic bacteria *Enterobacter cloacae*, *Enterococcus faecalis* provide a readily transferable tool for insect rearing systems that improve fitness by providing additional protection against infection, or by normalizing insect physiology increasing survival rates[42].
Beyond this SoA, RECOVER will explore different insect larvae in addition to that reported in literature for their capability to degrade plastic mixtures and also targets to improve plastic-degrading capabilities of insect larvae by strategies involving optimization of rearing conditions, including feeding or mixing with microorganisms. Improved insect larvae are mainly intended for ex-situ treatment of AWP in insect rearing chambers (in WP4) and, after their death, the adults produced will be used for chitin extraction in WP5.

C. State of the art: EoL plastic monitoring.

The sorting of polymers is a key step for the recycling industry. Equipment to automatically sort plastics are already widely commercially available (e.g. via spectroscopic identification systems, including near-infrared, X-ray, colour, or laser, and combinations thereof). All these identification techniques require a pre-calibration. Among those, near infra-red (NIR) spectroscopy is a well-established technique for inline composition monitoring. It is indeed highly sensitive to the chemical signature of a wide range of materials, especially organic ones. Using advanced data analysis approaches to treat the resulting spectra, it can predict fine multi-parametric information on the concentration of the substances of interest. Furthermore, hyperspectral imaging (HSI) is chemical imaging technique that records a hyperspectral image of the sample, providing not only the local chemical composition, but also its distribution. As such, the use of hyperspectral cameras goes far beyond the standard camera-based vision systems more widely spread in the industry. The spatial localization of the sample is especially exploited for sorting, e.g. plastic waste in the recycling industry, but also for analyzing the homogeneity of a product surface or identifying the presence of foreign bodies based on both chemical and physical insights. NIR and HSI were for example proved to be able to detect the presence of pesticides residues down to ppm levels[43]. While the use of HSI to identify plastics is quite common, its use to track plastic disintegration/biodegradation is rarely reported in the literature, some related data were mainly found to the related microplastics detection in ocean water[44,35,45].
RECOVER will demonstrate the capacity of using HSI optimized for each purpose below in terms of optical hardware and of data treatment algorithms (calibration thanks chemiometrics or even more sophisticated Artificial Intelligence approaches such as neural networks) at different steps of the process. This to automate the: i) identification of input End of Life materials composition (possibly used in the future composting plants as a pre-sorting, sorting of the materials they should feed into the degradation reactor); ii) detection of toxic substances that could jeopardize the later degradation such as pesticides which could be found on mulching films; iii) tracking of the degradation in terms of plastic fragments size reduction along time for each type of plastics as a support to later degradation optimization (to reach this objective, strategies to maximize the penetration of the HSI analysis in the compost will be devised and/or ability to relate the surface analysis to bulk composition will be checked).

D. State of the art: Biodegradation modelling.

Even for conventional biodegradable plastics, biodegradation is poorly controlled and understood mechanism influenced by the crystallinity, crosslinking, molecular weight, additives, but also by the environment (moisture, temperature, pH, microbial activity…). While different models, such as BIOWIN, BESS, METABOL, etc. exist for the prediction of the biodegradation of organic compounds (chemicals), **no model is available for the prediction of the biodegradation of polymers.** Due to the underlying complexity and number of influencing variables, and even more when talking about novel conventionally non-biodegradable plastics degradation, it seems unrealistic to envisage a mechanistic model that would properly describe the actual behaviour of such plastic item in our target new end of life process conditions. In contrast, data driven approaches can cope with such challenges. Data representing decomposition profiles can be modelled by benchmarking different machine learning algorithms accounting for the effects of the known parameters influencing the disintegration and biodegradation. "Deterministic" controllable test parameters and properties (polymer type, exact vermicompost composition...) and the less controllable

41 Brandon, A. M., Gao, S. H., Tian, R., Ning, D., Yang, S. S., Zhou, J., ... & Criddle, C. S. (2018). Biodegradation of polyethylene and plastic mixtures in mealworms (larvae of Tenebrio molitor) and effects on the gut microbiome. Environmental science & technology, 52(11), 6526-6533
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43 Optical Technologies for Determination of Pesticide Residue Yankun Peng, Yongyu Li and Jingjing Chen, Chapter 24 in Infrared Spectroscopy – Materials Science, Engineering and Technology, Theo M. Theophanides, InTech, 2012 - 510 pages, ISBN 953510537X, 9789535105374
44 Shan et al., Simple and Rapid Detection of Microplastics in Seawater using Hyperspectral Imaging Technology, November 2018, AnalyticaChimicaActa 1050, DOI:10.1016/j.aca.2018.11.008
45 https://daim.idi.ntnu.no/masteroppgaver/014/14972/masteroppgave.pdf

environment characteristics (moisture, temperature, microbial activity, etc.) can be differentiated. A Case Based Reasoning (CBR) approach can also be adopted in which successful configurations (i.e. environmental parameters and compositions) are stored in a historical database. CBR has been successfully used for relevant bio-industrial applications such as fermentation[46].

Beyond the state of the art lacking any model for polymer biodegradation and building upon the BBI project Biontop that started in June 2019 focusing, among other, on modeling the biodegradation in industrial and home composting conditions of novel polylactic acid (PLA) copolymers, in RECOVER, data driven models will be developed to predict the degradation based on the results of the tests performed with different plastics and enzymes/microorganisms/insects mixtures in WP4. The resulting predictive models will be embedded in a decision support system (app) to provide recommendation on the most suitable process depending on the AWP composition to recover as monitored in WP1 for the upscaled trials and future exploitation stage.

E. State of the Art: added value products based on chitin and chitosan

Extraction

Chitin is a polysacharide that can be found in the exoskeletons of invertebrates, but also in fungi, green algae and yeast cell walls[47]. Chitosan has degree of deacetylation (DD) of at least 60%, and is generally described in terms of DD and average molecular weight[48].Only chitosan with an average molecular mass of 10^5 to 5×10^6 g/mol shows antimicrobial activity[49,50,51], currently chitosan with a similar molecular weight is considered in packaging research [52,53,54].

Chitin can be extracted by classical chemical methods or by biological approaches. Either case require the exoskeletons to be cleaned, dried and ground into a powder prior to the actual chitin extraction by deproteinization, demineralization and final conversion by deacetylation. In chemical extraction, demineralization is conducted under acidic conditions whereas deproteinization and deacetylation is performed under alkaline conditions. Even irradiation can be used for deacetylation[55]. Chemical extraction is effective and feasible at large scale, but harbors several disadvantages: by-products of chitin extraction such as minerals and proteins cannot be used for further applications and it is environment-unfriendly as high amounts of waste and wastewater accumulate. Biological approaches applying either enzymes (not economically viable), or fermentation by starter cultures[56] have been proposed, considering that several bacterial species are applicable for chitin extraction and conversion to chitosan[57]. [58]. Biological methods favors the further usage of chitin extraction by-products, but is microbiologically complex, depicts incomplete process yield and takes longer than chemical methods.

When irradiating polymeric materials application of lower doses enhances cross-linking of the material whereas higher doses lead to material degradation. The irradiation of raw material accelerates downstream steps during conventional chemical extraction[59]. This clearly indicates that irradiation is able to initiate degradation of chitin.

Beyond this state of the art, in RECOVER we aim to establish a novel method for chitin extraction on laboratory scale using, irradiation by electron beam (eBeam) treatment to degrade the tight chitin network allowing the application of recyclable and more sustainable solvents for the actual chitin extraction. Additionally, eBeam treatment will convert α-chitin to chitosan, and reduce the amount of alkaline solutions and acids during conventional chemical extraction. Moreover, extracted chitosan will be evaluated regarding DD level and average molecular mass to meet the requirements for high performance and (active) packaging.

Chitin and chitosan processing for coatings and films

Chitosan is characterized by non-toxicity, biodegradability, film-forming capacity, antimicrobial and antioxidant properties and

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good barrier properties of films against oxygen[60,61,62]. **By chitosan application, it is possible to realize films and coatings with intrinsic antimicrobial properties and this is just this characteristic that differentiates chitosan from other common antimicrobials (for example, ethanol, sorbic acid, bacteriocins as nisins, lysozyme, essential oils etc.).**
Antimicrobial and barrier properties depend on chitosan molecular weight and deacetylation degree, its concentration, the solvent used for its solubilisation, the pH and possible plasticizers or other additives added in the formulations. Antimicrobial activity is possibly owing to the positive charges which can interact with negatively charged residues of macromolecules on the microbial cell surface, causing membrane leakage[63].
The thermogravimetric analysis (TGA) of chitin shows a weight loss between 250 °C and 400 °C that indicates deacetylation and cleavage of glycosidic bonds, chitin is therefore much more stable than starch and can be processed together with thermoplastic polymers by extrusion. Thus, PLA and chitosan can be processed by extrusion[64] as experienced by UNIPI on laboratory scale, and on pilot scale in the activity performed in NANO-CHITOPACK and POLYBIOSKIN project. As well several biodegradable PLA/PBAT based films are available on the market and their processing was experienced by RECOVER partners such as the formula patented by UNIPI for production of flexible transparent films based on PLA and a reactive plasticizer[65] .
Coatings based on chitosan may be not transparent when printed. Since in some packaging, the transparency of the films is required; this issue will be addressed in the optimisation planned in the project since based on literature[66] even in presence of chitosan transparent films may be achieved, the issue is mainly focused in achieving a good compatibility in the blends that can be obtained with food grade approved additives such as poly(vinyl alcohol)[67] and polyethylene-graft-maleic anhydride[68] (used in amount lower than 4% not to hinder compost ability).
Beyond the state of the art compounds based on compostable bio-polyesters will be processed for films production in presence of chitin/chitosan, as well commercial compostable or soil biodegradable films will be used as substrate for coatings with chitosan for active packaging and improved mulching films due to chitin or chitosan presence as well as for agriculture items (pots, yarns, etc), thus in addition to direct effects on plant nutrition and plant growth stimulation.

Chitin-derived products have also been shown to be toxic to plant pests and pathogens, induce plant defences and stimulate the growth and activity of beneficial microbes. Chitin-based treatments augment and amplify the action of beneficial chitinolytic microbes. In addition to direct effects on plant nutrition and plant growth stimulation, chitin-derived products have also been shown to be toxic to plant pests and pathogens, induce plant defences and stimulate the growth and activity of beneficial microbes, since chitin-based treatments augment and amplify the action of beneficial chitinolytic microbes69.
Plastic mulching (PM) is an emerging agricultural practice as it provides several beneficial properties. The use of PM is known to facilitate pest[70] control by suppression of weed growth[71,] and to reduce the amount of applied herbicides[72]. Moreover, PM functions as light regulating[73] greenhouse, increases the soil temperature[74,] and reduces evaporation that conserves the soil humidity[75]. Therefore, the agricultural yield is higher[76] and harvest in cool climate[77] is facilitated which provides an advantage in the market

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competition with warmer countries.
Market volume worldwide of plastic mulches is estimated to be around 700.000T per year just for LLDPE[78] and over 1 MTons of mulching films are used worldwide[79,] but also shows an impressive growth rate of about 25 % with enormous potential for assimilating new mulching technology[80.] PM is mostly based on LDPE, and suppresses gas exchange[81] resulting in changes in soil air composition. This can negatively affect soil quality and gas microclimate as oxygen deficiency may occur. A variety of strategies has been discussed and are partially already under application, e.g. strategies for temperature, moisture and gas exchange regulation under plastic mulches by the use of differently-colour and perforated films [82] as well as strategies optimizing soil conditions including green manuring,[83] planting a vegetation cover between mulch-covered row[84,] and using phytochemicals[85] or conventional fungicides.
Beyond this state of the art, RECOVER will modify existing conventional PMs by application of by-products from AWP degradation and chitin extraction. We aim to optimize gas permeability of PMs by modification based on determined permselectivity of the applied components for optimized conditions of the soil provided by the mulching films. The utilization of chitin extraction byproducts in optimized mulching films provides significantly increased added-value and closes the loop in agreement with a full biorefinery approach.

Irradiation to enhance Chitin/chitosan bioplastic products
Irradiation curing is applied in various industries to enhance the polymerization and adhesion of different types of coatings to distinct substrates as reviewed by Mehnert[86], irradiation treatment for crosslinking of monomers on natural films induce a significant improvement of the physico-chemical properties of films, improving the compatibility of the polymer network and water resistance[87].
For several polymers including chitosan blends irradiation at low doses can enhance barrier function as well as the mechanical properties[88,89] given that cross-linking is the predominant effect. Moreover, it was demonstrated that irradiated chitosan has a higher antimicrobial activity than unirradiated chitosan[90,91.] It was demonstrated that low dose eBeam irradiation of chitosan prior to coating additionally enhances the preservation. i.e. inhibition of fungal growth, and further prolongs the shelf life[92].
Beyond this state of art in RECOVER the plastic substrates coated with chitin or chitosan in WP5, will be irradiated to improve the mechanical and barrier properties. By performing low dose electron beam irradiation curing on wet and semi-dry chitosan coatings, RECOVER will further enhance the performance of chitosan.

F. State of the art: added-value products. Biofertilizers and biological control agents.
A bio-fertilizer is substance which contains living microorganisms that promote plant growth by increasing the supply or availability of nutrients to the host plant, including all organic resources for plant growth which are rendered in available form for plant absorption through microorganisms, or plant associations, or interactions[93]. The commercial history of bio-fertilizer began with the

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launch of "Nitragin" by Nobbe and Hilther in 1895. This was followed by the discovery of *Azotobacter* and then Blue-green algae and a host of other microorganisms which are being used till date as bio-fertilizer[94]. Bio-fertilizers are usually amended with carrier material to increase effectiveness of the bio-fertilizers, they are environment friendly and do not cause pollution unlike inorganic fertilizers which often 'run off' into water bodies causing eutrophication and 'blue baby syndrome. Bio-fertilizer acts as a soil conditioner adding organic matter to the soil which helps to bind the soil particles together preventing soil eructing, desertification, and erosion while increasing the water retention capacity of the soil[95]. It enriches the soil with beneficial microorganisms while boosting the already existing ones unlike chemical inorganic fertilizers which acidify the soil making it hard for microorganisms to survive. Organisms that are commonly used as bio-fertilizers components include nitrogen fixers (N – fixers), potassium solubilizers (K – solubilizer) phosphorus solubilizer (P – solubilizer), phosphorus mobilizers (P – mobilizers), used solely or in combination with of fungi[96]. The most important limitation of bio-fertilizer is their nutrient content when compared to inorganic fertilizers, that can be curbed by the addition of substances such as bone meal, wood ash or organic wastes. New amendments should still be explored to improve performance of biofertilizers available in the market. Among the components currently employed for biofertilizers formulation, chitosan stands for one of the most emerging ones utilized in the induction of the defense system in both pre and post-harvest fruits and vegetables against fungi, bacteria, viruses, and other abiotic stresses. Besides that, chitosan has been employed in soil as a plant nutrient and has shown great efficacy in combination with other industrial fertilizers without affecting the soil's beneficial microbes. Based on exhibiting such excellent properties, there is a striking interest in using chitosan biopolymers in agriculture systems[97].

Besides the use of more sustainable fertilizers the moving toward a "conscious agriculture" is relying on the biological control (BC) to replace pesticides. Biological control is the use of a population of one organism to reduce the population of another organism. Four different types of biological control are known: natural, conservation, classical, and augmentative biological control. Natural biological control is an ecosystem service whereby pest organisms are reduced by naturally occurring beneficial organisms. Conservation biological control consists of human actions that protect and stimulate the performance of naturally occurring natural enemies. In classical biological control, natural enemies are collected in an exploration area and then released in areas where the pest is invasive, often resulting in permanent pest population reduction and enormous economic benefits. In augmentative biological control (ABC), natural enemies (parasitoids, predators or micro-organisms) are mass-reared for release in large numbers either to obtain immediate control of pests in crops with a short production cycle (inundative biological control) or for control of pests during several generations in crops with a long production cycle (seasonal inoculative biological control). Biological control has been in use for at least 2000 years, but modern use started at the end of the nineteenth century. Cock et al. [98] mentioned 170 species of invertebrate biological control agents that have been used in BC in Europe while Van Lenteren[99] provided a list of about 230 species of invertebrate biological control agents that have been used in pest management worldwide, but recognised that this list was not yet complete. After predators and parasitoids, microbial biological control agents are the next most commonly used organisms in BC. About 209 microbial strains from 94 different species are currently commercially available for control of pests, but the numbers may be larger. During the first decade of the twenty-first century fewer new biological control agents came to the market, but during the second decade we again experienced a new phase with strong growth in both the development of new agents and a market for biological control[100.]

Beyond the SoA, following a cascade valorisation strategy, RECOVER will explore new biofertilizers based on chitin/chitosan extracted from insects and biological control agents. Such bio-fertilizers will play a key role in productivity and sustainability of soil and protect the environment as eco-friendly and cost effective inputs for the farmers.

Innovation potential

The innovation potential from the RECOVER project derives from its main outcomes:

- Advancement in biotechnology and enzymes selection
- A selection of high powered microorganism selected and specialised in mineralization of AWP
- A selection of worms and insect able to assimilate and degrade AWP
- A valorisation of RECOVER by-products such as vermicompost, fertilizers derived by worms/ insect, and chitin/chitosan extracted from insects into active packaging and enhanced mulching films.
- Additional environmental services (bioremediation, LCA, TEA assessment, process design, etc)

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Advancement in biotechnology and enzymes

Biotechnological improvements related to AWP management is a cutting-edge issue in worldwide R&D projects in order to produce novel approaches including several EU & International programs. Promising results from several groups using enzymes, microorganisms and specially insects and worms are increasingly obtained. In that respects, RECOVER will deliver new microorganisms and enzymes capable to produce plastic degradation, obtained by isolation from i) plastic polluted environments, ii) selected insect/earthworms with these abilities, or iii) neosynthesized enzymes developed using *in silico* technologies. Several advances to incorporate exogenous enzymatic capacities on substrates such as biochar are right now included in cutting-edge research institutions vg. USDA and others[101] to be used as active surface to reduce pollution by immobilization and/or degradation[102]

High powered microorganisms selected and specialised in mineralization of AWP

AWP presented in agricultural soils are commonly a complex mix of compounds with significant differences in terms of degradability. The identification of new microorganisms and specially the combination of microorganisms expressing novel or improved enzymatic activities enabling the degradation of mixtures of plastics is a major goal in the innovation potential of RECOVER. In addition, the development of formulations based on the efficiency tested in ex-situ and specially in-situ conditions will be a significant output.

Earthworms and insects able to assimilate and degradate AWP

Increased research dealing earthworm and insects abilities to degrade xenobiotics and plastics are reported. However, no significant data are found in literature about the combination of enzyme-microorganism and earthworm/insect consortium for magnification of AWP degradation. RECOVER propose break-through procedures to produce biodynamic platforms using earthworm and insects as symbiont with incorporated consortia of functional microorganisms/enzymes to be used in diffuse media such as soils or natural to enhance or produce active bioremediation of specific ions, compounds, nano or microparticles. This is a significant outcome to be produced in order to obtain specific functionality on biotic or abiotic platforms or combined systems. These complex systems including insect metabolism and specially earthworm mucus can produce also a long-lasting cleaning capacity far over performing any other enzymatic approaches. In terms of plastics EoL management, RECOVER will reduce the operation costs and environmental impacts of conventional AWP treatments based on in situ degradation.

Valorisation of RECOVER by-products such as vermicompost, fertilizers derived by worms/ insect, and chitin/chitosan extracted from insects into active packaging and enhanced mulching films.

Worldwide and EU fertilizer market are right now in a new reorientation on the product formulations due to the polluting actions on hydrosphere and atmosphere of conventional fertilizers vg. reactive Nitrogen (N), Phosphorus (P) overdose etc. The new approaches are in line with Circular Economy portfolio, looking to biobased actions focused on microorganisms, including biofertilizing, biostimulant and other added value properties including depolluting effects vg. Pesticides or other hazards such as microplastics. This process will be leaded by joint venture between administration, policy-makers and ground-breaking innovation, including a potential market of >1000 million euro. RECOVER formulations are designed to act not only on biofertilizing purposes but also on specific uses such as biocontrol and bio stimulating capacities by the inclusion of beneficial microorganisms-enzymes activities by RECOVER procedures. RECOVER will also act as a NPK major nutrient recovery strategy from waste producing slow release fertilizers avoiding the hazards linked to direct application of raw waste fluxes and contributing also to reduce the emergent pollutants (antibiotics, plastics and pesticides) increasingly appearing in fertilizers and composts[103]. The use of RECOVER solutions (composting, insects and earthworm consortia) will produce a virtuous synergy to enhanced process, enhanced quality of derived fertilizers (higher fertilizer capacities, lower salts contents, higher humic like compounds, and specially the reduction of plastic particles that will be considered in next compost quality assessment vg. microplastics).

Chitosan-derived products may prove useful as replacements raw material and as brand new components for: nanotechnology, as substitute advanced materials for plastics in consumer products, as carriers and encapsulators of microorganisms (biotechnology), biosensors raw materials, for certain medical applications, etc. Production of chitin/chitosan can be done by several procedures, such as chemical and biological extraction. Concerning insects as chitin/chitosan source, results of comparative analysis demonstrated that the quality of chitosan from insects is as good as from shrimp shells which shows that insects are indeed an effective, inexpensive alternative chitosan source which is readily available[104,105,106]. In addition, powdered

101 Ro, 2016. Bioenergy Res. 9:447:453
102 Sanchez-Hernandez et al. 2018 Sci Total Environ. 612: 1407-1416
103 Compost quality and safety for agriculture Summary report, WRAP, 2017
104 Badawy, R.M., Mohamed, H.I., 2015. Chitin extraction, composition of different six insect species and their comparable characteristics with that of the shrimp. Journal of American Science 11, 127–134.
105 Liu et al. 2012. Extraction and Characterization of Chitin from the Beetle Holotrichia parallela Motschulsky. Molecules 17, 4604–4611..
106 Marei, N.H., El-Samie, E.A., Salah, T., Saad, G.R., Elwahy, A.H.M., 2016. Isolation and characterization of chitosan from different local insects in Egypt. International Journal of Biological Macromolecules 82, 871–877. https://doi.org/10.1016/j.ijbiomac.2015.10.024

chitosan from insects is available for industrial, environmental, cosmetics and pharmaceutical purposes[107].

Additional environmental services

The coupling of AWP degradation with the resource production of biodegradable plastic can be a driving force to produce a virtuous cleaning circle providing ecosystemic services to modern agriculture and urban waste management.

RECOVER procedures can be used to produce abiotical products with exogenous enzymes to enhance bioreactions in agricultural soils to reduce GHG emissions, enhance N atmospheric fixation etc. Significant investments on cleaning of agricultural soils all over the world from high-refractance pesticides or its degradation products due to the increased standards of production of worldwide food distributors (trace-free labelling in food). RECOVER insects may have another commercial output within the sector of biological pest control. In two areas: "in vivo" production of natural enemies in biofactories, which can reduce production costs; its use as "non-pests" food in the open air and in greenhouses to improve and increase the management systems of insect pest natural enemies acting as the biological control (with production of natural enemies), an exponential increased market (valued 40 million euros per year only in Almeria province (Spain)[108]. RECOVER earthworm not only will degrade AWP but also can produce during this process an enhanced carbon sink effect due to the use/deposition of AWP microparticles from the topsoil in the burrow wall, obtaining an additional barrier to food chain transfer. The use of living biodynamic platforms (*Lumbricus terrestris*) will add an additional virtuous cascade of sustainability by improving physical and hydro-physical properties in intensivist soils, removing plastic debris from radicular zone.

Augmentative biological control is largely a market-driven enterprise with a defined, but constantly changing, economic value. It is estimated that approximately 230 natural enemies are available from over 500 suppliers with an end-user market value >$200 million globally (2012 data)[109]. In 2015, the global pesticide market had a value of US$ 58.46 billion[110]. The global market of biological control agents (invertebrates and microorganisms) was approximately US$ 1.7 billion in 2015[111], which is less than 2% of the pesticide market. Growth of the market for synthetic pesticides is expected to be between 5 and 6% over the next five years[112], but interestingly, growth of the biological control market has been faster: it showed an annual increase of more than 15% per year since 2005. The largest European biological control companies are still getting the main part of their income from sales of invertebrate biological control agents, but the contribution of microbial biological control agents is steadily increasing. Commercial ABC is used in protected crops and high-value outdoor crops, contributing to about 80% of the market value of invertebrate biological control agents. The remaining 20% of the market value for natural enemies comes from application of relatively simple, cheap but effective biological control programs often using only one biological control agent (e.g., *Trichogramma* spp. against lepidopterans in cereals and sugarcane, and *Cotesia* spp. against lepidopterans in sugarcane). Almost 40% of the income of the European companies originates from invertebrate biological control agents sold for control of Thrips, another 30% for control of whitefly, 12% for control of spider mites, 8% for control of aphids, and the remaining 10% for control of various other pests. Since 2005, predatory mites have contributed enormously to the growth of the market for invertebrate biological control agents as a result of: the (re)discovery of their use for control of whiteflies, finding more efficient species for Thrips control, the development of techniques to enhance dispersal and establishment of predatory mites in crops, and the development of new highly economic production technologies.

RECOVER can produce environmental solutions on waste treatment and valorisation, one of the most significant area to innovate in order to achieve win-win strategies in industries with increased costs linked to environmental assessment plans. In urban solid waste management, the enhanced treatments must be applied mainly on non-separated rubbish, a significant concern in non-developed countries. Nowadays in China, the use of insect-based urban waste management facilities is increasing. Shandong Qiaobin Agricultural Technology is right now managing 50 m^3/day using cockroaches in Jinan city. On wastewater fluxes, other RECOVER outcomes could be linked to the use of specific activated earthworms and insects in vadose polluted zones to obtain bio-restoration capacities, vg. pesticides, dyes, water bodies with high BOD/COD contents.

As seen, the innovation potential of the RECOVER technologies is extremely broad in the area that will be explored and beyond, but to ensure we can reach it, we have also carried a patent analysis which will be much reinforced in the project WP7.

107 https://ento.farm/; http://sflyproteins.com/sfly-products/
108 Teruel-Moreno M. (2010). El sector del control biológico en España. Informes y monografías no. 33. Fundación Cajamar, Almeria, Spain
109 Naranjo SE, Ellsworth PC, Frisvold G (2015) Economic value of biological control in integrated pest management of managed plant systems. Annual Review of Entomology 60: 621-645.
110 Research and Markets (2016a) Global pesticides market segmented by type, application area and geography. Trends and forecasts (2015-2020). Sustainability, regulation & competition. http://www.researchandmarkets.com/research/4hd338/global_pesticides
111 Dunham, Evolution and future of biocontrol. Paper presented at the 10th Annual Biocontrol Industry Meeting (ABIM), Basel, Switzerland, October 20th, 2015.
112 Research and Markets (2016a) Global pesticides market segmented by type, application area and geography. Trends and forecasts (2015-2020). Sustainability, regulation & competition. http://www.researchandmarkets.com/research/4hd338/global_pesticides

Freedom to operate

The use of insect *Tenebrio mollitor* for degrading waste plastics and recycled plastics, and bacterial strains capable of degrading petroleum-based plastics, and microbial consortia including such strains has been already patented[113]. Also, enzymes for the degradation of PET[114] and method for recycling mixed PET plastics involving enzymes[115] has been developed by CARBIOS and patented (but in that case the enzymes are encapsulated and compounded in the plastics rather than added in EoL management). In addition a method of degrading a plastic by the use of the fungi *Aspergillus oryzae* or *Aspergillus sojae* and also transformant microorganism recombined with DNA containing a gene encoding a surface active substance has been developed by Tohoku Techno Arch Co., Ltd[116]. However up to date there are no patents on the use of earthworms, other insect species and novel enzymes acting in a wider range of plastics, and even less on their innovative combination as planned in the project, therefore we are confident we have freedom to operate and even to protect our breakthrough approaches.

Whole value chain coverage

As previously mentioned in terms of multidisciplinary considerations, as well as in link with the interest to exploit the different above discussed processes and products to be delivered in the project, the RECOVER consortium, complemented by its AB, covers the whole required bio-based value chain to bring the project results to the market. The actions for the new value chain set up are described in WP7 and its foundation and different segments further justified in section 2.1 in link with the call topic KPI2. In particular, we cover the supply of raw materials for the new biotech process (EoL plastics can be provided by IDELUX, ASAJA and SAV; and NATURPLAS and FEMTO, respectively as post consumers and post-industrial wastes), enzymes (ASA), insects (NUTRINSECT) for the bioprocess. We also count with equipment and technology suppliers to optimally perform the biodegradation (OWS RF in terms of composting and AD reactors design, IRIS in terms of process monitoring and modeling). Finally, in view of the circularity of our value chain, the end-users of the RECOVER process partly overlap with the raw materials suppliers to recycle. Indeed ASAJA and SAV can use it to remediate the soil contamination by mulching films and generate potentially even a new business through, as for IDELUX, the treatment of their wastes. The initial plastics converters can in turn become the users of the chitosan extracted as a by-product from the AWP valorisation: FEMTO and NATURPLAS can benefit from bio-enhanced packaging materials and mulching films respectively. Keeping in closed loop and in view of a cascade valorisation, the by products from the biotech process developed are also used by the agri-food sector end users: improved food products shelf life (MANOR) and improved fertilizers composition (ENTO and SAV).

2. Impact

2.1 Expected Impacts

EC has presented a roadmap to support the transition to a knowledge-based post-petroleum economy in EU by establishing sustainable and competitive bio-based industries and value chains[117]. The Bioeconomy Strategy and its Action Plan pave the way to a more innovative, resource-efficient and competitive EU society, while generating new incomes and ensuring job creation and accelerate progress towards the UN Sustainable Development Goals (SDGs)[118].

In this context, the BBI's SIRA aims to push, among other things, the **development of new biotech processes and biomaterials helping decoupling economic growth from resource depletion and environmental impact**. Nowadays, plastics are used in various sectors with a demand of 50 Mtons per year; plasticulture accounts for 4% (greenhouse 31%, mulch 45% and silage 24%); while packaging makes up for 40% of this use, of which food packaging accounts for around 90%. In both sectors, the EoL treatment of plastics is underperforming. According to PlasticsEurope[119], in 2018, 31% of plastic waste was recycled in Europe but landfill and incineration are still dominant treatments for plastic post-consumer waste. The recently adopted EC's plastics strategy aims at decoupling plastics production from virgin materials, reducing their leakage into the environment, improving the economics and uptake of recycling to support the transition from a linear to a circular economy model.

RECOVER is a high impact project for the improvement of the bioeconomy in Europe. Thus, it is hinged on delivering biotechnological tools applying microorganisms, enzymes, earthworms and insects to degrade conventional non-recyclable plastic packaging and agricultural films waste streams but also result in new feedstocks (chitin/chitosan-based plastics and coatings and new fertilizers) for the primary sector and bio-based food industries. In this way, the circular economy concept is realized, providing added EoL for plastics which are currently landfilled or incinerated, or more often

113 Yang, Jun; Yang, Yu; Wu, Weimin. Biodegradation of petroleum-based plastics. U.S. Patent Application No 14/440,045, 3 Sept. 2015. US20150247018A1
114 Boisart, Cédric; Maille, Emmanuel. Method for recycling plastic products. U.S. Patent Application No 14/443,524, 15 Oct. 2015 WO2014079844A1
115 Maille, Emmanuel. Process of recycling mixed pet plastic articles. U.S. Patent Application No 15/310,458, 27 Apr. 2017. WO2015173265A1
116 Abe, Keietsu, et al. Method of degrading plastic and process for producing useful substance using the same. U.S. Patent Application No 10/532,423, 18 May 2006. US20060106120A1
117 COM(2012) 60 final. The Commission Communication of 13 February 2012 entitled "Innovating for Sustainable Growth: A Bioeconomy for Europe".
118 https://sustainabledevelopment.un.org/sdgs
119 Plastics – the Facts 2018. An analysis of European plastics production, demand and waste data

left on the soil. This impact will be achieved by performing research work with the objective of making this concept commercially viable in practice, analyzing the potentially successful exploitation pathways and assessing and managing any related risks.
Importantly, RECOVER will contribute to the achievement of the SDGs as discussed below in each relevant impact point and visualized in the Fig. 2.1. As further shown in this figure, RECOVER significantly contributes to the KPIs set out in the overall BBI-JU roadmap and more specifically to the expected impacts in the topic *BBI-2019-SO2-R3 Apply microorganisms and/or enzymes to resolve end-of-life issues of plastics*, as described below:

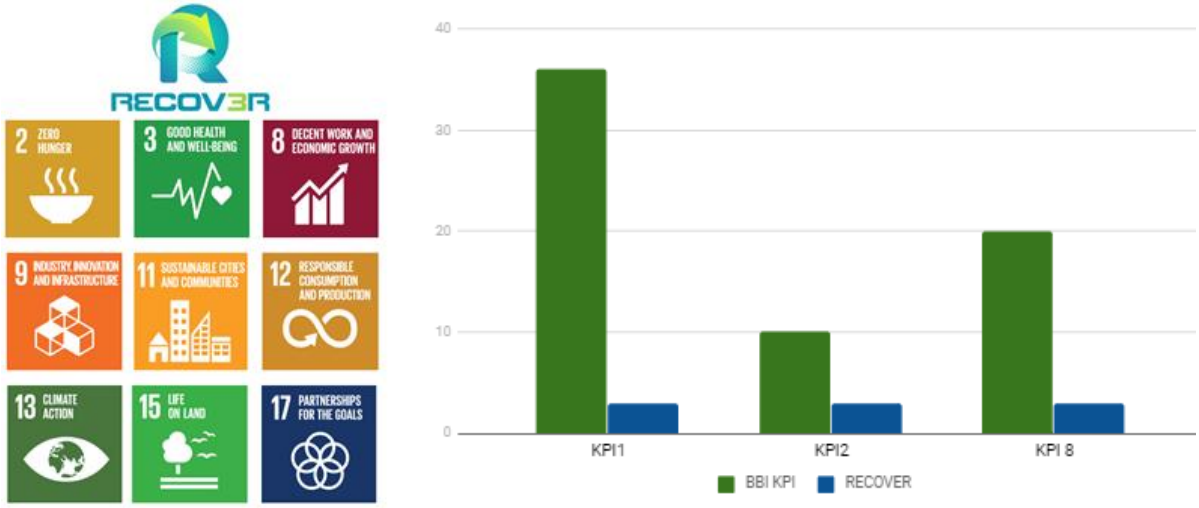

Figure 2.1: RECOVER contributions to SDGs and BBI JU KPIs

EXPECTED IMPACTS LINKED TO BBI JU KPIS:

▪ contribute to KPI 1 – create at least one new cross-sector interconnection in the bio-based economy;

The RECOVER proposal builds upon the BBI goal of achieving the full potential of the bio-based industries, which requires swift and concerted action of today's dispersed stakeholders across various sectors, disciplines and Member States.
In RECOVER, connecting different partners, such as researchers, technology developers and producers, through clustering and networking is key to developing new value chains and supporting the emergence of affordable EoL solutions for plastics, which also constitutes a strong contribution to the SDG 17 "Partnerships for the Goals". The active involvement of all key players along the value chains (including additional sectorial associations in the Advisory Board) will allow us to create long-lasting links with related clusters, necessary to fill a clear gap in the marketplace for sustainable plastics and EoL thereof. With RECOVER we will establish 1 new circular- cross-sector connection and further intensify 2 connections, among waste management and biotechnologies, and among the bioplastics, packaging/agriculture and agri-food sectors as explained below.
RECOVER will create an unprecedented systematic, cross-border and cross-sectorial, circular connection between all actors for the considered end of life management of non-biodegradable AWP. The new cross-sector interconnection is realised bonding the biotechnology providers, namely the suppliers of enzymes (for the plastics degradation, represented in our consortium by ASA) and insects (for the plastics decomposition, represented by NUTRINSECT) with users (post consumers AWP: IDELUX, ASAJA, SAV and post-industrial AWP: NATURPLAS, FEMTO) and converters (OWS RF for composting & biogas plant design, IDELUX for MSW management) of conventional fossil-based plastics. Following such approach RECOVER will not only allow to achieving high levels of biodegradation of AWP - both from fields or landfills - but also achieving its valorisation through insect by-products - such as high value chitin-chitosan-based formulations from the conversion of AWP. These last- will result in bio-fertilizers, new active mulching films and agriculture products (pots, sticks, yarns) for the primary sector as well as in active packaging for food industries.
The magnitude of RECOVER new cross-sector interconnection will be multiplied by ASAJA as top member of COPA and SAV, whose presence in the consortium favours the generation of positive spill over effects among its affiliates leading to push-pull effects for the new RECOVER symbiotic bioprocesses.
Relying on the transition from linear to circular value chains, we close the loop making farmers and food producers that generates AWP, become producer of biobased, biodegradable active ingredients for bioplastics (chitin/chitosan) as well as end users of new fertilizer formulations and mulching films, pots etc. with enhanced properties by AWP biodegradation by products. This way we will create new upstream links with producers of unavoidable AWP opening up unprecedented business opportunities for farmers that conventionally are positioned at the end of such value chain. This opportunity reinforces the genuine urgent need of the farmers to find a viable way to handle plastic pollution in soil for their sector to be sustainable on the long term. Indeed, RECOVER will create new incomes for agri-food sectors not yet involved in waste management to actually tap into the opportunity of converting their waste into value (more details in the paragraph related to impacts associated to the creation of new incomes). As such, the current waste challenge turns into an enabler for innovation allowing a significant decrease in the environmental impact of the agri-food sectors and of the value chains of the RECOVER solutions. We will have the agri-food sectors playing a strategic

polyvalent end user role within our bio-based value chain. While, as seen, they act as material supplier along the new value chains, they need to be supplied with agricultural films as well as fertilizers (both as insects by products, as chitin/chitosan based formulations). Incontrovertibly, this magnifies the strength of the new links and the resonance of our solutions in the new circular bio-based value chains. RECOVER will further **intensify 2 emerging cross-sectorial interconnections in bio-based economy cluster:**

- We will **reinforce the bond that the bioplastics sector has with the packaging sector** (which is by far the largest consumer of biodegradable and bio-based plastics, expected to grow at a CAGR of 17% to 2024[120]) by providing new demonstrators in food applications, possible thanks to RECOVER solutions' unique EoL tailored to their needs. This new link is for example illustrated at the level of our Large packaging converter, FEMTO, and food packer MANOR.
- We will **intensify the interconnection that the bioplastics sector has with the agriculture sector** (which is expected to grow at a CAGR of 5.8% to 2024[121]) by providing new demonstrators in agriculture films, pots, sticks, etc. applications. This new bond is realized through the Agricultural Film Supplier NATURPLAS and FEMTO producer of plastic items by injection moulding. The participation in RECOVER will ensure NATURPLAS to drive the development towards actual market needs and new legislative constraints.

As seen, the sectors embraced by RECOVER are of outmost significance in terms of need for new EoL with over 43% of plastic used for packaging and for agriculture, but the new cross-sectorial interdependencies created, considered also when building our AB, can contribute to the outmost exploitation of our results, also in other target sectors where plastics have difficult EoL (e.g. consumer goods, tableware, textiles, sanitary items, etc)..

▪ contribute to KPI 2 – set the basis for at least one new bio-based value chain;

The RECOVER consortium mobilizes the necessary partners and sectorial approach to contribute to the achievement of the KPI 2 aimed at delivering "10 new bio-based value chains by 2020". The development of a suite of novel biotech tools for the AWP biodegradation that respond to the demand for affordable and sustainable EoL solutions for plastics, requires liaising a critical mass of actors that are able to generate value working in a synergistic collaboration**.**

As such, RECOVER consortium leverages partners at every stage of the supply and value chain, as well as additional stakeholders in the AB.

In RECOVER, we create **3 new stable and effective value chains** not only by new players interconnections at the upstream of the chain (to link standard plastics with insects - microorganisms and enzymes providers) but also remodeling existing value chains that will substitute their petrol-derived products with our bio-based solutions (biofertilizers, agricultural bio-films and food packaging), as illustrated in Fig. 2.2.

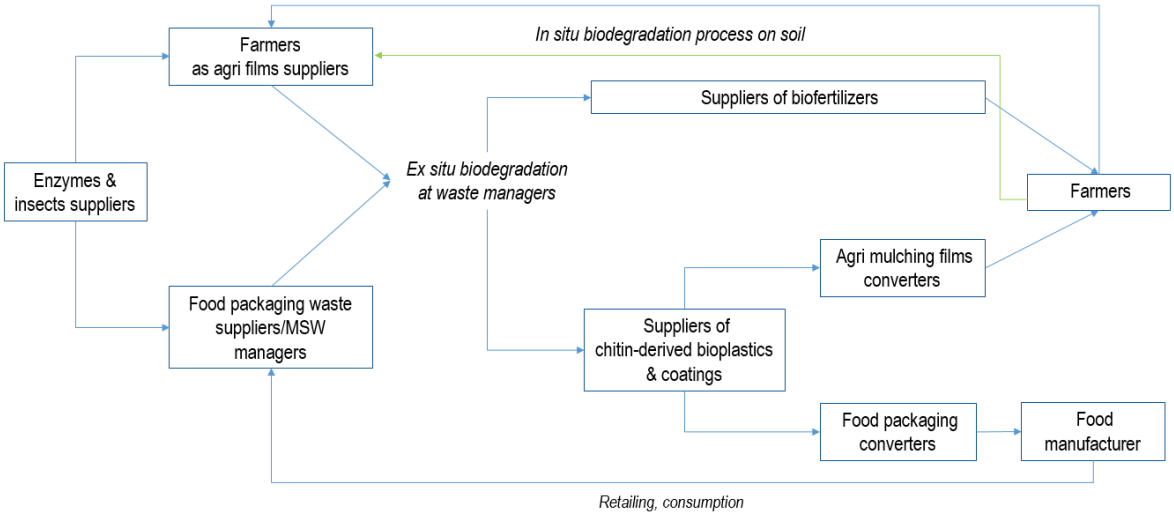

Fig. 2.2 RECOVER value chains

RECOVER is uniting expertise in biotechnology, plastic processing technology, biodegradability, LCA, engineering, as well as industry actors at different levels of the value chain, capable of steering the research in tune with their market knowledge and requirements. To reach the project impact of the delivery of bio-technology solutions for plastics EoL requirements that are currently not satisfied and that represent a very huge challenge for the environment, joint actions will be needed at different levels of the value chain, e.g. between farmers/municipalities, insects producers, material producers, final item designer and waste managing body so the biodegradation is actual. The *in* and *ex situ* biodegradation models will be a common tool used at these different levels facilitating this collaboration.

The value chain approach will also guarantee that access to selected knowledge occurs among the partners to reach a

120 The Future of Bioplastics for Packaging to 2024. Smithers Pira
121 Agricultural Films Market - Europe Market Shares, Trends and Forecasts (2017 - 2022). Mordor Intelligence

seamless exploitation. The RECOVER actors will work towards similar goals and a proportion of Large end users in the consortium or AB, who will test and validate the new solutions in the selected target demonstrator applications, are included as early adopters at the end of the value chain to stimulate the demand. This will particularly benefit SMEs (IRIS, ASA, NUTRINSECT, ENTO, NATURPLAS) as it will provide them with new up- and down-stream partners. Cooperation among RECOVER members will be particularly important in the post-project phase, as results of the project are pre-competitive and additional investment will be required before RECOVER is ready for commercialization. As addressed through specific tasks in WP7, setting the basis for a robust value chain will be key to generate real value on the market. Building on the experience of strong players in the composting/bioremediation sectors will play a massive role in designing promising solutions for plastic degradation and shaping their successful market entry phase. As such, we are confident that our developments, meeting true market demand, can convert into a sustainable value chain.

▪ contribute to KPI 8 – validate at least one new and improved processing technology reflecting the 'TRL gain' since the start of the project.

(1)

(2)

(3)

Fig. 2.3: Equipment for the 3 mentioned processes showing increased scales potentially used along the project where applicable *(resp. process (1) at OWS RF, process (a) at ASA and process (3) at ASU)*

As discussed in TRL table 1.4, we will follow a scale up approach of the different technologies involved in the project (generally from TRL 2-3 now to TRL 4-5 at project end). In terms of the most relevant processing technologies for the biobased industries for which we will provide advances that pave the way towards future industrial replication, we would like to stress 3 cases:

1- The **ex situ composting of the AWP**: it relies on SoA at TRL3 mainly and will be incrementally brought to TRL5. In that aim, tests will be made in test tubes and small vessels (up to 1L, TRL 4) in WP2 and 3 and then upscaled to reactors of 50 L at OWS RF and IDELUX (fig 2.3).

2-For the first time **AWP-degrading enzymes and microorganisms** will be produced in quantities suitable for the treatment of plastics. Indeed, the production of these enzymes and microorganisms will be upscaled from shake flasks up to 200 L fermentation volume thanks to the participation of ASA (fig. 2.3). The in silico studies will greatly support this work.

3-the **extraction of chitosan** from chitin will be brought from TRL 2 to TRL4 when referring to the combined use eBeam irradiation and green solvents for that purpose. This approach is aimed to facilitate the solubility and suitability of chitin towards the application of less harmful and/or bio-degradable solvents either directly through eBeam treatment or by eBeam mediated coupling of moieties. eBeam treatment of raw material with low doses (20 and 25 kGy) accelerates downstream steps (deproteinization, demineralization, and deacetylation) during conventional chemical extraction that aim, ASU will extract in vessels of increasing sizes (up to 2L) in WP5 (using the same eBeam equipment along the upscale of the eBeam assisted extraction, fig. 2.3). This is a highly innovative process that is much more sustainable and safer that the current ones employing alkaline solvents and promises to attract a lot of interest from the biotech industry.

ENVIRONMENTAL IMPACTS:

▪ decrease the amount of non-biodegradable polymers sent to disposal or even discharged to the environment;

RECOVER will provide novel biotechnological solutions applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films wastes.

Agro-plastics

As seen in the *Ambition part, "A. State of the art and advances in the field of plastic controlled waste management and treatment technologies"*, the majority of the AWP in Europe consists of LDPE (low density polyethylene), EVA (ethylene-vinyl acetate) and copolymers for mulching films; Spun-bonded non-woven PP (polypropylene) fabric, LDPE and EVA for direct covers; LLDPE,

LDPE, EVA and PVC (polyvinylchloride) for low tunnel films; everyone of them for greenhouses and PE (polystyrene) and PP (polypropylene) for fertilizers sacks.

The EU agriculture and horticulture industries generate 1.200 tons of plastic waste per year, around 9% of which is recycled, 12% incinerated, and 79% was accumulated in agro-fields (buried), landfills or the natural environment[122]. If current production and waste management trends continue, projecting current global waste management trends to 2050, 12,000 Mtons will be discarded or buried in the natural environment and 12,000 Mtons incinerated producing high levels of CO_2 emissions[123]. In addition, because of the durability of the polymers involved, substantial quantities of discarded end-of-life plastics are accumulating as debris in landfills and in natural habitats worldwide. The agricultural grounds covered by greenhouse for protecting cultivation account for 1,300,000 ha globally. Of the 80 Ktons of mulch films marketed per year in EU, only 5% are biodegradable. Agricultural plastic wastes constitute an environmental and economic problem, because they may cause severe and diffuse damages to the landscape, the environment, the agricultural soil, the air (in case of incineration) and to the shallow and deep water if they are not properly managed. Plastics that are used for agriculture is typically buried, abandoned along rivers, or in rural areas or burnt in open air often on farm property or uncontrolled sites. Plastic films generate about 80% of the agricultural plastic waste. For these applications, the plastic waste collection and recycling are difficult or even impossible due to the extreme amount of soil and crop fractions therein. The contamination by soil, sand, and organic material of plastic agro-textile materials can reach up to 60-80 % of the total amount in terms of weight, making the mechanical recycling economically unviable. This unavoidable plastic accumulation is soil has led to a demand for soil biodegradability of plastics impossible to collect and recycle.

Non-recyclable packaging

Packaging may contain high traces of food and other residues that complicate a standard recycling requiring cleaning and separation pre-treatments prior to reprocessing. The complex nature of the plastic waste streams in terms of polymer composition and the presence of multilayer films and other flexible fractions difficult to handle aggravates the issue. Several technical, waste stream handling and cost efficiency problems do not allow the wider expansion of recycling.

In RECOVER, the optimised biodegradation process will be performed either ex-situ in composting reactors and insect rearing chambers (preferred route for non-recyclable plastics from municipal solid waste or for agricultural films that can be collected easily) or in-situ in the case of bioremediating soil pollution, by e.g. mulching films, further to contributing to the long-term removal of non-biodegradable polymers from the environment.

Both, enzymes and microorganisms have been reported as successful in biodegradation of LDPE, HDPE and PS among others fossil plastics. Likewise, the insects used in the process explained above are reported as highly trained for the ingestion, digestion and conversion of AWP. Therefore, AWP biodegradation by successful communities of enzymes and microorganisms will produce sub-products that insects will take advantage of as food. Taking into account the life cycles of the insects used in this procedure, we will be able to tune them to ingest AWP, digest it and turn it into chitin, we can also plan in which period of their life cycle they have a better performance. In this way and taking into account the reported results we can achieve the expected level of biodegradation and even much higher. One of the most disruptive contributions will be the use and enhancement of a natural capacity (in the larval stage) of several of the insects used: Larvae can significantly accelerate degradation by altering the structure and abundance of their associated bacterial community. This quality associated with biodegradation under common environmental conditions will allow us achieving the expected levels of biodegradation, even in conditions economically sustainable.

As seen in contribution to KPI 1, in RECOVER we contemplate the valorisation of the process by-products to develop bio-enhanced mulches with a clear link to the criteria of the upcoming European standard for the biodegradation (through RECOVER process) of plastic mulch films in soil. Once the use of agricultural films ends, no removal from the field for disposal transporting them to incineration ovens (PE or PP based) or industrial composting facilities (PLA) will be required providing vermicompost to be applied on the soil. This can reduce the agricultural plastic waste accumulation and gain no toxic effects for soil and crops. So the controlled biodegradation in soil is considered as a sustainable solution for agro-films and indeed a very high impact innovation as opposed to landfilling or above described uncontrolled disposal approaches leading often to littering.

Thus, the RECOVER bioremediation solution as the deriving bio-based and biodegradable plastics will bring enormous advantages in avoiding the generation and dispersion of microplastics in our fields through i) the soil bioremediation, ii) ex-situ valorisation and iii) the insurance that compost from currently collected organic fraction often polluted by microplastics is clean. As an additional output of the bio-remediation process, we envisage the development of biofertilizers, that, instead of contaminating the ground, they will fertilize it for future crops. As such, RECOVER is a concrete contribution to the achievement of the SDGs and specifically to “SDG 2: End hunger, achieve food security and improved nutrition, and promote sustainable agriculture”; and “SDG 15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss”.

Biobased biodegradable chitin/chitosan packaging and compounds will favour the market penetration of bio-based plastic for specific applications.

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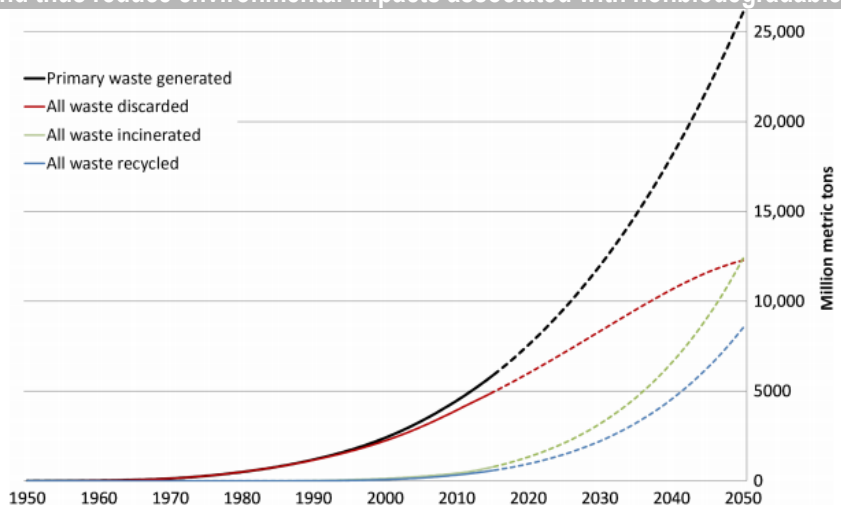
122 Briassoulis, D., Babou, E., Hiskakis, M., Scarascia, G., Picuno, P., Guarde, D., Dejean, C., 2013. Review, mapping and analysis of the agricultural plastic waste generation and consolidation in Europe. Waste Management & Research, 31(12): 1262-1278

123 R. Geyer, J. R. Jambeck, K. L. Law, Production, use, and fate of all plastics ever made. Sci. Adv. 3, e1700782 (2017).

Considering the RECOVER target applications the proposed AWP and plastic packaging waste valorising affects 23% (half of the packaging being the flexible and/or multilayer fractions and all agri-films) of all the plastic produced yearly (65Mtons in EU in 2017)[124]. **RECOVER proposed organic recycling solutions could, in a long-term scenario, tap upon all these agri films and in the range of 50% of the currently not-recyclable packaging in MSW (in view of other emerging solutions for recycling complex fraction, we consider here a limited part, e.g. when very fouled or not well pre-sorted).** Whereas only the 31% of the plastics plastic are currently recycled in EU, RECOVER could **potentially increase by ca. 12% the plastic recycling in EU** (7,8Mtons of AWP, which will no longer be landfilled neither incinerated, but rather valorised).

▪ in case the focus is on the removal of non-biodegradable polymers from the environment, contribute to 'close the loop' and thus reduce environmental impacts associated with nonbiodegradable polymers discarded in the environment;



The vast majority of monomers used to make plastics, such as ethylene and propylene, are derived from fossil hydrocarbons**. None of the commonly used plastics are biodegradable. As a result, they accumulate, rather than decompose in landfills or the natural environment. The only way to permanently eliminate plastic waste is by destructive thermal treatment, such as combustion or pyrolysis. Thus, near-permanent contamination of the natural environment with plastic waste is a growing concern.** The largest groups in total non-fiber plastics production are PE (36%), PP (21%), followed by PET, PUR, and PS (<10% each).

Fig. 2.4 Cumulative plastic waste generation and disposal (in M tons)[125]

RECOVER is hinged on the removal of fossil-based plastics used in agricultural and packaging sectors and specifically on PE, PET, PP, PS, EVA, and laminates which are non-biodegradable and currently have a very low recycling rate (due to difficult recyclability in view of their fouling and /or complex structures). In RECOVER, we provide value added EoL solutions for plastics which are currently landfilled or incinerated (with a cost), or left on the soil, applying microorganisms, enzymes and insects to degrade conventional plastic packaging and agricultural films waste. Microplastic contamination on land is estimated to be between 4 to 32 times higher than in the oceans. In addition to inadequate EoL treatment of plastic waste, plastics reaches our soils through increasing use for agricultural purposes. Yearly inputs of microplastics in European farmlands are estimated to be 63,000-43,000 tons.

All in all, RECOVER proposes two alternatives for removing the amount of non-biodegradable plastics from the environment: i) the in situ bioremediation of the agricultural plastics left on the soil applying vermicompost based on a tuned combination of insects, enzymes and microorganisms (such an approach could also in the long term be adjusted as contingency for accidental littering in highly contaminated areas); ii) microplastics removal from compost, a critical problem for the current organic waste management companies, due to the presence of non-biodegradable plastic contaminants non properly sorted in the supposedly organic fraction collected from e.g. MSW. As such, **RECOVER will strongly contribute to the achievement of the "SDG 3:** Ensure healthy lives and promote well-being for all at all ages", **by avoiding air, water and soil pollution with** hazardous **fossil-based plastics and indirectly to "SDGG 11: Make cities inclusive, safe, resilient and sustainable" in the case of AWP from MSW.**

Such ways of removing plastic let RECOVER to implement circular bioeconomy principles as AWP will be converted into biofertilizers from insect manure and chitin extraction residues improved by the incorporation of PMC, fortified worms and insects/larvae that will have biopesticide, biofertilization and biostimulation effects, thus closing the loop within the agricultural sector. In addition, the project will deliver chitin/chitosan extracted from the vectorized insects after their use which can in turn be a new source of bioplastics used both in agriculture (packaging, sticks, pots), and food packaging (rigid containers, films) contributing to establish a circular economy paradigm in our target sectors. Therefore, RECOVER will be a key enabler for implementing the Circular Economy Package (CEP). Specifically, RECOVER contribute to the implementation directive on the packaging and packaging waste, which sets new legally binding targets and fixed deadlines for waste recycling and the reduction of landfilling. Indeed, the package includes a common EU target for recycling 65% of packaging waste by 2025, and 70% by 2030; in this context, the plastic sector has a separate target 50% by 2025 and 55% by 2030. This represents a significant increase from the current target imposed by the Packaging Directive 94/62/EC, which set a 22.5% recycling target.

Savings in transport will be also considered since the very beginning of the project by logistic studies, for plastic waste collection. In

124 Plastics – the Facts 2018. An analysis of European plastics production, demand and waste data. Plastics Europe (2018)
125 Geyer, R., Jambeck, J.R., Law, K.L. (2017):Production, use, and fate of all plastics ever made. Sci. Adv. 2017;3

that sense, RECOVER provides affordable and sustainable solutions to contribute to the achievement of the ""**SDG 13: Take urgent action to combat climate change and its impact.**

▪ enlarge the range of feedstock able to be effectively and sustainably processed through biocatalytic systems.

Current biotechnological processes use biomass as main feedstock, thus the use of synthetic plastics is a significant change of paradigm. Building on recent literature showing this possibility, in RECOVER we develop novel biocatalytic systems (microbial communities, novel enzymes and mixtures/symbionts of them with insects and earthworms) that will enlarge the range of feedstock processed for the production of chitin-based bioplastics and fertilizers covering multiple types of synthetic plastics (e.g. PET, PS, LDPE etc.) in AWP streams. We contribute with several specific and innovative communities of microorganisms, novel enzymes, insects and earthworms with improved and novel AWP degrading capabilities whose biocatalytic performance for processing these challenging streams will be optimised in pilot reactors. The LCA results will be considered to support the most sustainable process to be upscaled.

The biotechnology-based solutions provided by RECOVER constitute an alternative approach in mitigating the challenges associated to plastic waste management and simultaneously provide novel pathways for the synthesis of biopolymers and new potential routes for the effective degradation of synthetic plastics. The environmental and ecological benefits of the biologically mediated degradation (i.e., mineralization) of plastics are meritorious on their own. In addition, **the biotechnology-based strategies proposed in RECOVER also contribute to further enhance biodegradation mechanisms in obtaining added value products**. Hence, plastic waste provide a new valuable resource that might contribute for the entry of plastic litter into a circular economy model[126]. As such, plastics will also be converted by biological processes where carbon will be returned to nature in a safe and sustainable way, e.g. by compost and biofertilizers.

As said, present bioproduction processes are primarily based on sugars or sugar-containing materials including organic wastes or lignocellulosic materials. From technological point of view they suffer at least from two major problems: 1) limited supply of sugar-based substrates for large-scale bioproduction and 2) high costs of available substrates (accounting for up to 50% of the total costs in many cases) and up- and downstream processing (nearly the other half of the total cost)[127]. In addition, in some cases, food competing feedstocks are used. Bio-based materials produced from biomass instead of fossil resources addresses GHG emissions reductions, central to the mitigation of climate change. But this requires more biomass, which can impinge on food security, and can interfere in many of the ways highlighted above. The interferences can partly be ameliorated by the use of synthetic plastic (as non-avoidable waste occurring at many levels in our society) for biotransformations as proposed here. With the help of improvements to plastic-degrading enzyme versatility, **RECOVER would allow enzyme technologies entering into biorefinery models and green chemistry.** The biopolymer indirectly generated from synthetic plastic degradation could then be used to make more biodegradable plastics, which would represent a completely biological recycling of plastics in a circular economy[128]. The integration of biocatalytic systems (hydrolytic enzymes, microbial, insects and earthworms) would result in a custom biocatalytic system platform that is capable of converting synthetic plastic into biodegradable counterparts in a single step.

In addition to the use of synthetic plastic as a new feedstock for bioplastic production, the use of insect larvae and earthworms to upcycle feedstock resources currently lost, not efficiently used in the food chain or acting as a protein-rich feed ingredient for the livestock and aquaculture sectors is envisaged. The use of AWP residual streams as a feeding substrate for insect larvae result into environmental benefits (lower GHG emissions and especially lower land use) compared to conventional protein-rich insect feed ingredients with a high environmental impact. Thus values between 2.8 and 31.2 kg CO_2-e and between 0.06 and 14.5 m^2 per kg of protein larvae fed on municipal waste and beet-pulp, respectively[129] The use of organic resources can, for example, result in a competition with food, feed, fuel, and fertiliser production for natural resources. Using residual streams with a limited application is therefore recommended to avoid this competition. This competition can also be reduced by using residual streams as efficiently as possible, for example, using the remaining material as fertiliser.

RECOVER, by converting non-degradable plastics into biodegradable plastics and fertiliser, offer a new feedstock and unconventional route for the production of bioplastics from synthetic plastics. Since sustainable consumption and production aims at "doing more and better with less", **RECOVER provides solutions to contribute to the achievement of the "SDG 12: Ensure sustainable consumption and production patterns.**

ECONOMIC IMPACTS:

▪ lay the basis to the further exploitation at higher scales of microorganisms and/or enzymes to convert currently challenging streams;

In RECOVER, the proposed biotech solutions rely on symbiotic bioprocesses between high powered microorganisms through microbiotic systems vectored by several insect species. Indeed, one of the main innovations is **the combination**

126 EuropeanCommission. (2018a). Factsheets on the strategy for plastics in a circular economy. Retrieved from https://ec.europa.eu/commission/publications/factsheets-european-strategy-plastics-circular-economy_en

127 Zeng, A. P. (2019). New bioproduction systems for chemicals and fuels: Needs and new development. Biotechnology advances 37: 508-518.

128 Narancic, T., & O'Connor, K. E. (2018). Plastic waste as a global challenge: are biodegradable plastics the answer to the plastic waste problem?. Microbiology, 165(2), 129-137.

129 R. Salomone, G. Saija, G. Mondello, A. Giannetto, S. Fasulo, D. Savastano Environmental impact of food waste bioconversion by insects: application of Life Cycle Assessment to process using Hermetia illucens J. Clean. Prod., 140 (2017), pp. 890-905

of endogenous and exogenous microorganisms and their inoculation on insects for achieving higher levels of biodegradation and direct conversion (at the same time) of AWP in insect product such as chitin-chitosan, which is recognized as high value as bioplastics raw material and other interesting industrial applications).
RECOVER will select microbial communities, novel synthetic enzymes, earthworms and insects for AWP biodegradation. Test will be performed in TRL5 scale in WP4. Furthermore, the participation of the biotech companies ASA and NUTRINSECT will act as a key enabler for ensuring that the production of tailored enzymes and insects for the degradation of AWP is already performed in sufficient scale in the project and the full scale production is viable. Further entities in our AB will also support those aspects.
The LCA and LCC analysis, and the setting of a logistic approach as well as TEA will help in a planning of a viable future roll out of the project results and will strengthen RECOVER business plan. SMEs and Large industries in RECOVER are ready to invest with their in-kind contribution to the project and with future investments since they are convinced of the high exploitation potential of RECOVER outcomes, with significant revenues on their market as detailed in following paragraph.
The exploitation of such scales of microorganisms and enzymes will be ensured by the establishment of the 3 new value chains, as seen in contribution to KPI 2. From a biotechnological perspective, symbiotic microorganisms constitute promising and mostly untapped sources for potential applications in medicine, bioremediation, industrial processes, and agriculture.

▪ increase income and business opportunities for stakeholders and actors (including primary producers) in the bio-based sectors, in particular in the waste management sector
In RECOVER, new business opportunities - and consequent new revenue streams - will be opened for all the actors along the value chain (visualized in the Fig. 2.2). At the upstream of the RECOVER value chains, the suppliers of enzymes, microorganisms and insects will have the possibility to expand their activities in the bioremediation market. A new report from BCC Research, "Bioremediation: Global Markets and Technologies to 2023," estimates that by 2023, the market will reach $186.3 billion, up from $91 billion in 2018, indicating a CAGR of 15.4%. In situ bioremediation is expected to see the largest market growth compared to ex situ bioremediation. From 2018 to 2023, in situ and ex situ bioremediation are expected to grow at CAGRs of 15.0% and 16.7%, respectively.
As a secondary route, the insects produced may be applied in the sector of biological pest control. The global market of biological control agents (invertebrates and microorganisms) was approximately US$ 1.7 billion in 2015, with a CAGR 15%.
Currently the agricultural sector is relying on the yields and prices of their main products for their subsistence. However, there is a large potential for diversification of their businesses by valorising AWP, which would allow farmers address new business opportunities. **The biotechnologies set up in RECOVER will not only provide solutions for the degradation of plastics but also opportunities for AWP valorisation offering farmers new outputs from their business by having access to new processing to be able to commercialize new products**. As seen, agricultural plastic waste streams are a huge environmental and economic burden for farmers. For example, in case mulching films, it is estimated that, depending on the thickness, as much as 68% of the mulching film is not removed from the field as removal is time-consuming (about 16 hours per hectare) and disposal expensive (up to € 300 per hectare)[130].

Table 2.1: Expected values from resulting downstream products

Ex situ biodegradation output	Downstream products	Market price (€/kg)	Expected income for farmers/agri-cooperative (€)
Chitin from insects/earthworms	Biofertilizers	5-30[131]	258.600[132]
Chitin/chitosan to be added in food packaging and agriculture products (mulch, pots etc) or as anti-microbial, anti pests coatings	Bioplastics	5-6	387.900[133]

World production of main nutrients (nitrogen, phosphorus and potassium) currently stands at 190 million tons, of which just over 110 million of nitrogen, 45 of phosphorus and 35 of potassium.
The global biofertilizers market was valued at USD 1,326.6 million in 2018 and is expected to grow at a CAGR of 13.4% during 2019-2025 period[134] The demand of biofertilizers has been stimulated by augmenting practice for organic farming, high requirement for soil organic matter and favourable regulatory structure in Europe. For example, in Italy the legislative decree 33 of 22 January 2018 definitively closed the door to synthetic chemical pesticides[135] since by 2 May 2020 it is estimated that up to 95% of agrochemicals, including biological pesticides, currently on free sale, will be outlawed. The European Union is continuously encouraging the use of biofertilizers by advising the farmers to optimize the application of chemical fertilizers, or replace them partly or completely with biofertilizers that are eco-friendly and also result in better economic return, given that biofertilizers are cost-effective. The chemical industry knows that it must quickly alert its R&D departments to develop new products that do not require registration and satisfy consumer demands that will be (and in part already are) products based on natural substances, including those derived from foods

130 https://docs.european-bioplastics.org/publications/OWS_Expert_statement_mulching_films.pdf
131 Acién et al (2012) Production cost of a real microalgae production plant and strategies to reduce it. Biotechnol Adv 30:1344–1353.
132 Assumptions: 30tons sold on Year 1 and and annual linear increase of 5%.
133 Assumptions: 15tons sold on Year 1 and and annual linear increase of 5%.
134 Biofertilizers Market. Global Analysis by Trends, Size, Share, Business Strategies, and Key Developments: 2019-2025. Adroit Market Research
135 https://agricommerciogardencenter.edagricole.it/agrofarmaci/agrofarmaci-biologici-scelta-obbligata/;

(such as chitin, chitosan, garlic or rapeseed oil) and commonly used for various uses, in any case not classifiable as chemical, toxic or harmful 'environment. In this contest RECOVER vermicompost, and biofertilizer based on RECOVER by-products (biomasses, chitin/chitosan) can cover a wide and important market share in innovative and sustainable biofertilizer.
The global chitosan market size was valued at USD 5.71 billion in 2018 and is anticipated to expand at a CAGR of 20.8% from 2019 to 2025 due to growing product application scope in water treatment, healthcare, biotechnology, animal food, agriculture, food and beverage. Currently, the majority of available chitin in the world comes from shrimp shells waste, which accounts for 150 billion tons annually around the world[136]. Chitosan in agriculture is used to coat seeds for conservation during storage, for fertilizer releasing systems, pesticide formulation, thus the potential to enter this market through the RECOVER added-value fertilizers is very high.
Chitosan has been proven to control numerous pre and postharvest diseases on various horticultural commodities. It has been reported that both soil and foliar plant pathogens fungal, bacterial and viral may be controlled by chitosan application. For some horticultural and ornamental commodities, chitosan increased harvested yield. As a nontoxic biodegradable material and elicitor, chitosan has the potential to become a new class of plant protectant, assisting towards the goal of sustainable agriculture.
Over the last decade, there has been an increasing interest by the food industry in the development of novel bio-based active films as food packaging materials, not only due to their natural origin, but also because they can enhance food safety and quality during storage. Chitosan is one of the most promising candidates thanks to the combination of its excellent film-forming properties and antimicrobial activities. Active chitosan-based films have been used to preserve various food products, especially fruits, fresh produce, and meat. As a functional biopolymer, chitosan has many intrinsic nutritional values and health-promoting bioactivities against many chronic diseases. Due to its broad-spectrum antimicrobial activities, chitosan has been widely studied as a food preservative to improve food quality and extend the shelf life of perishable food products. Furthermore, chitosan has been approved as a food additive as in Korea and Japan since 1995 and 1983, respectively.
For the bioplastics, a recent market analysis forecasts the global bioplastic market to grow at a CAGR close to 20% over the period 2018-2024[137], certainly benefitting to RECOVER related downstream products. 70% of bioplastics currently produced are used in packaging and 5% in agriculture and horticulture.
RECOVER, by applying these technologies and approaches, will be producing a major positive impact on the primary sector which is responsible for nearly €200 billion on the EU economy[138]. To release this potential to the market, based on European market data relating to AWP, as well as data on the current and future potential of the bioeconomy, the project will identify new business models. In case of agricultural films that can be easily collected, in RECOVER the ex-situ composting reactor will be set up. As described in the Section 2.2, the total investment for running the RECOVER solutions is assumed to be almost €200.000. While large agricultural companies can afford significant capital and operational expenditure, traditional Micro and SMEs cannot. To increase their competitiveness, we aim to establish for novel collaborative business models based on sharing facilities, where farmers cooperating with other actors along the value chain, that result in increased flexibility and innovation capacity and new incomes for the primary producers. Thus, RECOVER will give impetus to innovative business models where such AWP valorisation can be offered as a service by agricultural cooperatives resulting in a creation of 'clusters of value' able to efficiently biodegrade AWP and solve the problem of plastic waste in their local fields. For this purpose, while shaping value chain in Task 7.2 *Roadmap for the set-up of RECOVER value chains and new cross-sector interconnections in the bio-based economy*, agri cooperatives members of COPA through ASAJA will be surveyed. In RECOVER, by products biocompounds from AWP valorisation offer a major source of renewably-sourced chemicals and bioplastics offering an enormous business opportunity, especially for fertilizers, packaging and agricultural materials which require biodegradability, and are key for a prosperous and competitive Europe. In case of bioremediation, i.e. in-situ biodegradation no new incomes will be generated for farmers, except for the cost savings for AWP disposal, which, as seen, worth about €300 per hectare.
To biodegrade conventional plastic packaging from MSW we consider that waste managers can use or replicate their composting reactors to degrade AWP and/or have already sufficient investment capacity to incur the costs for additional RECOVER composting reactor. As such they can act as first actor to bring in short the solution to commercial scale and they can also play a strategic role in the transition towards the creation of the agricultural value chains. Indeed, while treating plastics waste they can also open their facilities for bio-degrading locally collected agricultural films and AWP, thus providing feasible and sustainable short-term solutions for farmers, considering even that the transport of AWP must be kept as short as possible to reduce impact and costs. For companies commercialising currently non-recyclable AWP, the ability to recover them will result is reduce fees in terms of landfilling and incineration (in the range of 200€ per ton), once again boosting the shift to a new EoL model supported by the whole value chain.
RECOVER, through its action contributes to the achievement of the "**SDG 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation",** to the extent that it makes the considered sectors more sustainable, enabling the development of new, sustainable integrated business models, based on circular solutions, working towards the end goal of becoming fossil carbon-free and achieving zero waste, also in the primary sector. Further, it contributes to the "**SDG 12: Ensure sustainable consumption and production patterns".**

136 Production of Chitin and Chitosan from Shrimp Shells, A.A.Khan (Atif), H.Shibata (Hiro), M.T.A.P.Kresnowati (Penia), S.L.Tai (Tai), Conceptual Process Design, Process Systems Engineering, DelftChemTech - Faculty of Applied Sciences Delft University of Technology,2001
137 Bioplastics Market by Type and Application: Global Opportunity Analysis and Industry Forecast, 2018-2024. Allied Market Research
138 Rural areas and the primary sector in the EU. https://ec.europa.eu/agriculture/sites/agriculture/files/statistics/facts-figures/eu-rural-areas-primary-sector.pdf

SOCIAL IMPACTS:

▪ create new job opportunities in the bio-based sector, particularly rural and/or urban areas.

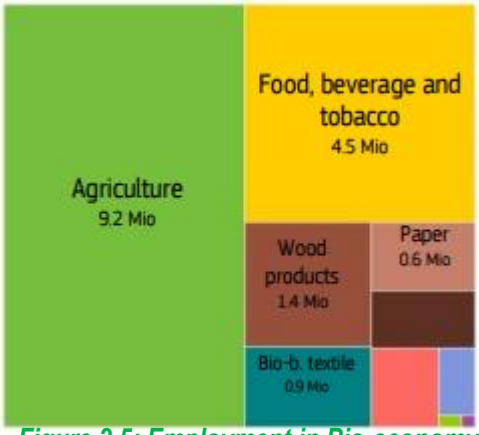


Figure 2.5: Employment in Bio-economy

The EU-28 bioeconomy employed 18 million people and generated €2.3 trillion of turnover or €620 million of value added in 2015. In other words, this sector employed 8.2% of the EU-28 labour force and generated 4.2% of the EU-28 GDP. Agriculture and the manufacture of food & beverages accounted for about two thirds of the value added and turnover of the bioeconomy and three quarters of bioeconomy employment (Fig. 2.5[139]). These sectors generated €174 billion and €233 billion of value added respectively in 2015. Agriculture, being a low labour productive sector, employs 51% of bioeconomy workers but generates only 28% of the bioeconomy value added. The role agriculture plays in rural areas and as an economic and social buffer is not to be underestimated.

In RECOVER, new jobs will be created both in rural areas as well as in urban areas, at all stage of the RECOVER value chains.

As seen above, from the upstream of the RECOVER value chains, new enzymes manufacturers and insects growers will increase their production as they will extend their business to the AWP degradation. For this purpose, they will require increasing job by 3.26% (valorisation activities in agri-wastes would create even much more jobs in rural areas).

In addition, the set-up of new incomes for farmers and farmers' cooperatives will help create jobs in rural areas, where land abandonment and jobs loss would have taken place. Indeed, as the RECOVER ex situ biodegradation will result in market valuable chitin/chitosan products, new revenue streams will derive for primary producers and expectedly new employment for them.

As set in the BBI roadmap, our proposal is expected to contribute to new skilled jobs in the biotech space: engineers, chemists, biologists, business experts, etc. as well as unskilled and semi-skilled jobs in product processing and related functions will be needed to operate the biodegradation plants.

Importantly, as the ex situ biodegradation solutions are proposed for existent landfills and other municipal waste management organizations, new incomes will be ensured for them which will be generated by the chitin/chitosan feedstocks to be converted, as seen in bioplastics used in improved mulching films and active packaging, as well as in new fertilizers thus, contributing to the creation of new job in urban areas.

All in all, based on a new job being created for every €200k of turnover, combining it with the expected revenue streams, outlined in Section 2.2, RECOVER is projected to create at least 3 new jobs for each farm organization/ agri-cooperative (as a transition scenario) and 7 for each MSW management organization in 5 year. In EU there are 220.000 agri-cooperatives[140] and 400.000 MSW landfills[141]. Estimating to reach a market penetration of 0.1% (totalling 220 agri-cooperatives and 400 MSW) in a 5-year period post project, i.e. horizon 2030, RECOVER will generate 660 new jobs in rural areas and 2.800 in urban areas (further linked to the expected revenues discussed in section 2.2[142]).

Estimates about how partners respectively expect to grow their business and create jobs with the new RECOVER solutions are given in section 2.2.

As part of action for job creation in the bio-economy sectors, we also work toward the job retention. As such, core skills and competencies will be assured along the value chain through the knowledge transfer, upskilling and training sessions that will take occur in RECOVER (*Subtask 7.4.4 Knowledge transfer & Upskilling).* This has to be seen as part of RECOVER contribution over the "**SDG4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all set the target".**

In addition, indirect measures will positively stimulate job creation and retention, resulting from the new business opportunities that will be created for all the actors along the 3 value chains in RECOVER as well as from the increased innovation capacity of the organisations.

As was discussed above, RECOVER will ensure that agriculture and urban waste management are brought into the innovation cycle in order to benefit from technological advancements and new knowledge. Thus, the coupling of AWP degradation with the resource production of biodegradable plastic is a net generator of decent jobs, and that it is also a vital strategy for our current and future generations. This move is expect to encourage municipal investment in AWP biodegrading plants and collection services, while also creating local jobs.

As such, RECOVER takes concrete actions to realize the **"SDG 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all".**

Other social and environmental impact

A key contribution that RECOVER will make to environmental and societal impacts is that of **decoupling agriculture and industry**

139 Ronzon et al. (2017). "Research Brief: Estimating jobs and wealth in the Bioeconomy". European Commission -Joint Research Centre.
140 https://ec.europa.eu/agriculture/sites/agriculture/files/agri-markets-task-force/contributions/concrete/copacogecaagricooperatives_en_.pdf
141 https://eurelco.org/infographic/
142 Based on a new job being created for every additional ˜€210k of turnover

growth and competitiveness from environmental impact. The economic development aspirations of our private industry sector are entirely compatible with reducing the impacts of AWP on the environment.
As seen, plastic waste accumulation in soil is not only detrimental to the environment (irreversible soil contamination, degradation of soil quality, pollution of rivers and groundwater) but also for human health (heavy metals, dioxins and furans released during plastic burning are associated with endocrine disruption, heart disease, cognitive and motor disabilities, and human carcinogen)[143].
In RECOVER, our efforts will contribute to consumer access to healthier soils and possibly seas and therefore food that contains less harmful substances leaking from plastics or microplastic fragments.
In terms of environmental impacts associated with non-biodegradable polymers discarded in the environment, **the RECOVER AWP-management could potentially avoid around 80% of CO_2 emissions caused by common incineration of plastic mixes**, when referred to the CO_2eq factor of emissions reported for the used microorganisms and macro-organisms in biodegradation (which could be neutralised by coupling it with a culture of microalgae as suggested by ALGAEENERGY in our AB).
This estimation derives from the reduction of the incineration ratio of AWP (12% of agri-plastics waste) and taking the average of low density polyethylene as the pattern of agri food plastics mixes. Counting on the CO_2eq production in incineration of polyethylene we could in an approximation obtain that AWP incineration produces 1,868 millions of CO_2 oil equivalent yearly (Institute for Energy Diversification and Saving (IDAE) data). This approximate will be much refined in the project LCA. AWP incineration is a partial combustion, and in itself generates a specific human and environmental mutagenic hazard (but not considerable for CO_2 calculation).
In terms of environmental footprint of EoL phase, according to the EC DG ENVI "Plastic waste in the environment"[144] study from 2011, plastic film recycling could potentially save 0.74 kg CO_2 eq./kg recycled plastic in comparison to landfill (>50**%). In RECOVER, we expect to reduce CO_2 footprint by at least 50%** (although uncontrolled waste management is not accounted for in LCA, and would in reality further contribute to an overall reduced environmental impact.
In terms of secondary raw materials from our valorisation process, bio-based plastics such as chitosan derivatives have the unique potential over conventional plastics for reducing GHG emissions or even being carbon neutral (1 metric ton of bioplastic generates between 0.8 and 3.2 fewer metric tons of CO_2 as compared to 1 metric ton of petroleum-based plastics)[145],[146].
Savings in transport will be also considered since the very beginning of the project by logistic studies, for plastic waste collection.
A beneficial CO_2 balance will be also achieved in RECOVER due to earthworms and insect growth
Positive impacts will derive also from the use of bio compounds, such as chitin and chitosan as fertilizer and pesticides. Indeed, biofertilizers will replace conventional chemical ones, reducing the environmental and health load associated to the use of chemicals in agriculture. In addition, chitin and chitosan compounds used in RECOVER for the active food packaging films allows longer shelf life of perishable food and result in less waste of food products, considering that the production of some perishable food such as meat has a high environmental load. The demand for our consumer products, RECOVER packaging, is expected to be driven by green- and health- conscious consumers concerned about the environment[147]. Furthermore, societal benefits will be assessed by the ASAJA that will give the alignment of RECOVER to the consumers' needs and perceptions and will contribute raising awareness about economic, societal and environmental challenges.

Contribution to improving the innovation capacity and the integration of new knowledge

Addressing our societal and environmental challenges, e.g. the end-of-life issues of plastics and the sub-optimal EoL compatibility of conventional plastic materials with existing waste management scenarios, is recognized to be a major catalyst for increasing our innovation capacity, and for the integration of new knowledge generated as a result of research into our practical societal and industrial contexts. This represents a unique opportunity for Europe to kick-start a world leading competitive circular bioeconomy with significant economic, environmental and societal benefits. The EC also unlocks tremendous potential for the development of innovative new materials and products derived from bio-based sources. In such context, the project will provide new knowledge on potential applications for biofertilizers, agricultural plastic application and food packaging. To increase the innovation capacity of the industry, our approach is to capitalise on a valuable opportunity for i) the EU farmers and municipalities, ii) packaging and fertilizers industries to respectively have access to i) novel sustainable, green and affordable solutions for biodegrading AWPs both ex and in situ and ii)continuous supply of biocompounds derived from chitin and chitosan for biocoatings and bioplastics as well as biofertilizers. The use of advanced approaches like online monitoring and process modeling in WP4, borrowed from advanced factories and industry 4.0 paradigms, will also enable farmers and waste manager to gain cutting edge knowledge in terms of their process.
While the generation of new knowledge is positive, it is in its industrial use that innovation capacity is increased, and impact is realised. As such, to allow generating truly market replicable innovation, the RECOVER consortium unites the industry value chain necessary for market success, and the project has integrated several cross-cutting activities such as environmental, regulatory, safety and economic assessments, and finally business planning activities to maximize the post-project exploitation of the results. The project gives trans-sectorial industry and transdisciplinary research and technology players an opportunity to collaborate and

143 D. Briassoulis et al, Review, mapping and analysis of the agricultural plastic waste generation and consolidation in Europe, Waste Management & Research, 2013, Vol 31, Issue 12, pp. 1262 – 1278
144 http://ec.europa.eu/environment/waste/studies/pdf/plastics.pdf
145 Fondazione Istituto Italiano di Tecnologia, Bioplastics and Bioelastomers
146ww.mckinsey.com/business-functions/sustainability-and-resource-productivity/our-insights/the-new-plastics-economy-rethinking-the-future-of-plastics
147 About 80% of European consumers want to buy products with a minimal impact on the environment (Eurobarometer survey, EC, 2013).

share all required technical knowledge and market expertise. Effective knowledge transfer from the RTDs (UNIPI, UAL, BRUNEL, UMH and ASU) to the participating enterprises, especially SMEs often lacking own R&D capacities, laying down the basis for supporting the industry partners to carry forward the results post-project, is fundamental to increasing the likelihood of market uptake. To this end, a dedicated task, T7.4.4 has been included in the work plan to ensure seamless knowledge transfer occurs.

Barriers/obstacles/framework conditions that may determine whether/to what extent expected impacts will be achieved

RECOVER is a very ambitious RIA set on achieving measurable results within a midterm implementation period and focused on creating a considerable legacy that will enlarge its impact beyond the project's duration. Below are described main barriers and obstacles that could hinder the expected impacts:

Economical barriers

The investment needed for transitioning towards the RECOVER plastics biodegradation solutions (CAPEX discussed in section 2.2) could be a barrier for market uptake but we estimate it will be effectively counterbalanced by the gain for avoiding incineration and landfilling taxes (e.g. 200€/Ton) and collection fees, that currently represents an economic burden for these non-sustainable EoL options.

Agricultural mulching films

Even with a new approach for soil remediation, it would be valuable for our solutions to meet not only the EN 13655 Plastics - Thermoplastic mulch films recoverable after use, for use in agriculture and horticulture: this is for conventional plastics that have to be recovered after use (chitin/chitosan presence will not represent an issue for incineration or for recycling due to the chemical structure and low amount that will be added in the films), but in the future target meeting the newly accepted standard EN 17033 Plastics – Biodegradable mulch films for use in agriculture and horticulture – Requirements and test methods (2016), for mulch film that can be incorporated into the soil due to their property of biodegradation, thus requires a 90% biodegradation and additionally some limitations on heavy metals and evaluation of toxicity.

Current PE mulching films have proven their efficiency for crop protection, but are not biodegradable in the environment, resulting often in undesired litter. Biodegradable mulch film can be a solution for the litter issue, but need to be very well developed to ensure that desired barrier properties are kept during use considering that each crop has different growth profiles and different environmental conditions. Therefore the effect of biodegradable plastic mulch in crop production with regards to microclimate modification, soil physical, chemical and bio-logical properties, soil moisture, weed control, soil nutrients, and pest and disease management needs to be carefully considered, making (apart from the higher cost) a market introduction more difficult compared to conventional mulching films. The chitin coating might help to improve barrier properties and expand the portfolio of crops, allowing the use for crops with slow growth.

Biofertilizers

Biofertilizers are not included in the legislation at a European level yet. In a close future, they are supposed to be regulated jointly with fertilizers by the European Union[148]. At the moment, we count with a European proposal Regulation 2016/0084 is to incentivize large scale production in the EU from domestic organics or secondary raw materials in line with the circular economy model, by transforming waste into nutrients for crops[149]. According to the proposal, the lack of harmonization in organic fertilizers prevents European companies an easy access to other State Members' markets. In comparison with inorganic fertilizers, where processes are already defined and harmonized, biofertilizers find important difficulties to extend its market in the EU. As a consequence, the sector cannot develop in countries with abundant manure to produce organic fertilizers but small expanses of crop, thus low demand. Non-defined and tough national legislations jointly with an obsolete European Regulation even for inorganic fertilizers from primary raw-materials (which render it impossible to keep up with the innovative cycle of the fertilizers sector), undermine the biofertilizers and biostimulants industries. In order to update the current Regulation and promote fertilizers based on second raw materials, in accordance with circular economy, the European Commission has provided a regulatory framework radically easing access to the internal markets for such fertilizers and addressing environmental concerns arising from contamination by EC-fertilizers of soil, inland waters, sea waters and food. For each of the fertilizing products and component materials, the proposal introduces specific harmonized requirements regarding quality (minimum nutrient/organic matter content), safety (maximum limits for heavy metal, organic contaminants, microbial contaminants and for impurities) and labelling. The proposal defines recovery rules for bio-waste transformed into compost as well as the use of CE marking for fertilizing products.

Chitosan coating for food contact

Our biobased active packaging materials fall within the scope of the Framework Regulation (EC) No 1935/2004, which provides the harmonized legal framework by setting out general principles and safety requirements for all kinds of food contact materials (FCM). As non-plastic FCM, chitosan is not yet covered by the harmonized legal framework. However, as it is regarded as active compound, the Regulation (EC) No 450/2009 is regarded as applicable. Moreover, Chitosan of animal origin must be in

148 Bourguignon, D. 2017. Briefing EU Legislation in Progress: CE marked fertilising products. Members' Research Service 1 December 2017, Fourth edition. European Parliamentary Research Service.

149 European Commission (2016). Proposal for a Regulation Laying Down Rules on the Making Available on the Market of Ce Marked Fertilising Products and Amending Regulations Technical Report EC 1069/2009 and EC1107/2009. Brussels: European Commission.

compliance with Regulation (EC) No 1069/2009 and Regulation (EU) No 142/2011. Regulation (EC) No 450/200 sets out rules on the composition of the active FCMs by establishing a positive list of substances approved for use in the manufacture of plastic FCMs ('Union list'). Compliance with regulatory demands is essential for putting new packaging materials on the market.
Also a possible reluctance may arise for the source of the chitosan, which derives from insects. However, in that regard it should again be noted that insect sub-products are already used in food production such as shellac in confectionery industry or Red Food Colouring additives. Moreover, chitosan from crustacean is already applied in dietary supplements.
Other possible barriers may arise in the case of in situ biodegradation from the farmers' fear to have the insects contaminating their surrounding fields. During our communication campaign, awareness raising efforts towards the message that this bioremediation solution will be run under strict control and with no contamination of adjacent agricultural crops. We will consider these aspects in WP7 regarding the social acceptance of our solutions.

Additional investments

In terms of further contributions to maximizing the impact of this BBI action, significant in-kind contributions are planned to reflect the industrial context of the action and will definitely contribute to increasing the impact and reaching the project objectives. The consortium covers in kind 23% (1.3M€) of the overall project budget. This co-funding, comprising the full budget of the large companies as well as a minimum of 10% for all others partners, is justified by the genuine conviction of the partners that they will on the long term obtain a return on their investment when the RECOVER solutions reach the market. Members of the AB who will make also significant in-kind contribution (in providing samples, assessment, advice, etc.). In addition, some partners also provide additional investments towards the BBI-JU's objectives (table 2.2, total >25M€).

Table 2.2: Summary and cost of the additional investments *(note: no financial transfer between partners are made)*

No	Partner	Activity	Contribution to the BBI JU objectives	Estimated private investment (M€)	Estimated value public investment (M€)	Other EU funding received
4	BRUNEL	A new Department of Chemical Engineering established in 2019, and is investing in new facilities and staff in bioprocessing and industry-focussed biotechnology, including enzymes.	KPI1, KPI2	-	0,5	NA
6	ASU	Innovation Campus – Model Factory and Laboratory for sustainable food production and food packaging	KPI1, KPI2 (polymer processing), KPI6 (food packaging)	2	1	NA
12	IDELUX	Investment in 2 new CHP engines of 885 kWe, the rehabilitation of the biowaste reception hall, a new pre-treated biowaste extraction conveyer belt, a new crusher feed hopper Rentec, and a new mixer axis of a pump (PA01)	Sustainable biowaste treatment and energy production through biogas and cogeneration	1.45 0.74 0.8 0.6 0.135 respectively	NA	NA
14	ENTO	Investment in equipment for Plant Production of Black Soldier Fly (Digestion capacity of 3000 tons of vegetable residues from industry and from commerce)	Vegetables from commerce "managed" in the plant as a residue or as source of carbon and enter the digestion. This must be tested out of plant and we need to have a test lab in INIAV (Portuguese National Institute for Veterinary and Agriculture Investigation).	18	NA	NA

2.2 Measures to maximise impacts

a) Dissemination and exploitation of results

Outline of the dissemination plan

The dissemination plan for RECOVER comprises a set of actions to spread the non-confidential results generated during the project to ensure that as many relevant audiences as possible have access to these results. The aim underpinning the dissemination strategy is to heighten awareness about the tangible and intangible outputs of RECOVER RIA, to contribute to knowledge building and advancement of innovation in Europe as well as to avoid duplicity of research work. In order to safeguard the commercial opportunity, the dissemination activities will subject to a confidentiality policy. By default, and building on top of the consortium experience in collaborative projects, any result of the project is to be treated as confidential and can be disseminated only after agreement of the project partners following the relevant clauses of the consortium agreement. This approach allows us to have the complete control and

make sure that no "exploitable" information is disclosed before being protected. The protection of the intellectual property rights (IPR) is of invaluable importance for the participating partners. Therefore, the non-confidential results and information on the project to be publicly disseminated will be carefully identified. The RTD participants in the consortium aim to publish results (and indeed, for us, publications are a very positive success indicator reinforcing the cutting edge aspect of our research and also serving as peer reviewed documents towards the industry, standards bodies and policy makers) while the industry partners and market actors are also eager to showcase and exhibit results at tradeshows and exhibitions and through other industrial channels (technical/sectorial magazines, trade fairs, etc.). The IPR & Exploitation Manager (Dr. Marco de la Feld from ENCO) will work closely with the project innovators to manage the identification of non-confidential disclosable information that can be published/disseminated. He will lead the project partners in strategically planning and effectively implementing the dissemination during a dedicated task in the Work Plan, *Task 7.4 Exploitation and Dissemination Planning*. All the measures (outlined in table 2.3) will be tailored taking into account the project needs, the target groups' background and the expertise of the project partners. Our project dissemination activities will intersect dissemination strategies of the BBI-JU, via BIC associated members in our consortium (UNI and UAL) and others such as SPIRE (IRIS, ENCO), which will contribute to the distribution of project outputs across the relevant European and international communities and amplify the project dissemination effort. All the dissemination activities will be developed with an eye towards the exploitation of the project results by other parties, in order to lay down the basis for subsequent innovation actions as well as other private initiative which can accelerate the route towards the take-up of new technologies within the plastic sector, also beyond the RECOVER selected applications.

We intend to approach both traditional channels such as dissemination at conferences (presentations, including submissions of papers and posters), publications in peer-reviewed and open access journals, dissemination in industry magazines, newsletters and online platforms, as well as more vanguard channels, such as social media platforms (ResearchGate, Academia.edu, SlideShare), project blogs, etc. Also the project website will serve as a lasting virtual hub for disseminating downloadable public deliverables and outputs of RECOVER, as it will house also the 'RECOVER Green Channel' for open access to the peer-reviewed publications of the project (it will be widely disseminated across Europe and worldwide to generate visitors and traffic to the site).

In particular, the network of COPA across the European agricultural value chain should be highly instrumental in identifying and approaching prospective business partners and potential licensees. With more than 60 farmers associations across the EU, ASAJA through COPA will echo the RECOVER results among the policy makers and major stakeholders in this industry. The association also has access to several channels of communication within the agricultural industry, ranging from the agri-food newsletter with the widest circulation, to one of the most widely-followed (12.5 k followers) Twitter accounts focusing on agri-food topics.

Table 2.3: Outline of the RECOVER Dissemination Plan

Target Audience	Tools & Channels	How contributing in achieving Impacts
Research community & Academic stakeholders	**OA Journal Articles:** Food Packaging and Shelf life (I.F.3.347); Food research International (I.F. 3.520); Innovative Food Science and Emerging technologies (I.F.3.116); Environmental Science & Technology (I.F.,7.149); Microbial Biotechnology (I.F.4.857); Bioresource Technology (I.F.5.807); Environmental Pollution (I.F. 5.714); Journal of Biotechnology (I.F.: 3.163), Nature Biotechnology (open access fee) (I.F.35.724), Microbial Genomics. **Specific events and symposia:** Innovation in Food Packaging, Shelf life and Food Safety Conference (DE); Open day at Universities, Researchers' Nights, Knowledge Open Festivals	Effective knowledge sharing among the research and scientific communities in order to accelerate the shift towards affordable solutions for biodegrading AWP
Early stage researchers		Attract more young people to scientific and research careers, enriching their curricula with knowledge from the project. Early stage researchers inspired by the results, will be motivated to remain in EU- research and possibly work in areas that are beneficial for society, sustainability and innovation for agrifood industry.
Farmers, organic farmers, farmer associations as well as films and plastic manufacturer for this sector	**Trade fairs/exhibitions:** BIOFACH (DE); SANA - International Exhibition of Natural Products - Nutrition, Health, Environment (IT); BIONORD - Organic Industry Fair (DE); BIOSÜD - Organic Industry Fair (DE); VIVEZ NATURE PARIS - Bio Agriculture and Natural Products Fair (France); Plastics Recycling Show Europe (Holland) **Magazines:** Farmers Weekly, Organic Farming Magazine (UK); Organic Life	Foster new sustainable, cooperative and inclusive agri-food value chains, replicating the RECOVER innovation in other territories and rural areas beyond the project boundaries. Multi-actor approach to support the transition towards a post-petroleum economy Leading to the wider adoption of RECOVER system across EU farmers.

Food processors **Agri-food enterprises,** **Packaging converters as well as plastic manufacturer for this sector**	**Trade exhibitions and conferences:** Interpack (GE); FoodTec&Anuga (DE); INNOV-AGRI (FR); Smart Packaging 2020 (GE); AGROKOMPLEX (SK); SEEDS and CHIPS exhibition; WorldBio, EFIB (EU), IFIB (IT), regional spearhead cluster meetings (FL), BIC/BBI meetings (EU);Bioplastics Business Breakfast (GE); International Conference and Exhibition on Food Processing & Technology (UK); Alimentaria (ES); Plastics News Europe Project's training events, PLAST2020 Milano, (IT) **Magazines/Journals:** Food Business News; Plastics News Europe; Food for life Newsletters: IFST newsletter	
Enzyme companies **Insects suppliers and earthworm** **Biological control companies**	**Trade fairs/exhibitions:** EFIB, Compounding World Expo, European Biotechnology Congress **Open Access Journals (OAJ):** Polymers,, Journal of Biotechnology, Nature Biotechnology, Microbial Genomics	Widespread of the use of the project innovation. Multiply success of the project by inspiring new initiatives based on the project results.
Fertilizers companies	**Trade fairs/exhibitions**: Agritechnica Hannover, Startup Europe Smart Agrifood Summit, **Open Access Journals (OAJ):** New bio-fertilizers, Green industries and valorization of agro-plastic wastes by high powered biodegradators.	Widespread of the use of the project innovation. Multiply success of the project by inspiring new initiatives based on the project results. Building new collaborative value chains among fertilizers clusters and associations.
Industry Multipliers- Sectorial Associations (PlasticsEurope, Fertilizers Europe, CEFIC, Plastics Europe, PAFA - Packing and Film Association, European Crop Protection Association, European Association of Agricultural Economists) and other associations	**Conferences, seminars and workshops** (participation in organising, scientific committees, with poster, papers, presentations, or chairing): Annual World Food and Innovation Forum; Bio-based World, Biopolpack, Biopol, Sustainable Bioplastics, Bioplastics, Biorefining, Bio-based Materials, Ecomondo, etc. **Features in Newsletters and e-bulletins:** food production daily, agronotizie, plastics today, food production daily, plastic news, etc.	Dissemination of new knowledge and technologies Building new collaborative value chains among food clusters
National food safety authorities, EFSA, EC, EIP Agri, FAO, WHO **Policy makers & Decision-makers**	**Policy dialogue:** joining roundtables fostering multilevel and horizontal discussion on RECOVER results. Collaboration with public administration will drive better informed decision making processes across Europe Include several requirements in PAC policies for awarding farmers for AWP collecting and valorising.	Market push Increase the influence of the project and its results in policymaking Creation of a positive framework for the take up of RECOVER results
H2020 projects, international/ national projects/initiatives	IoF2020, AGRICHEMWHEY, ANTARES, PROTEIN, STEP-IN, REFUCOAT, BioBarr, HYPERBIOCOAT, SuperBIO, BioSTEP, Open-Bio, InnoNet, BIOCOMPLACK, ReTAPP AGRIMAX, SHEALTHY, etc.	Engagement and cross-fertilization with other similar H2020 projects and reuse between previous and future projects.

Initial Exploitation and Business Plans for RECOVER Key Exploitable Results (KERs)

The RECOVER consortium mobilizes the necessary partners not only for the upscaling of the biotechnological solutions for degrading plastics (WP3), the scale up of both in situ and ex situ degradation trials (WP4) but also for the development of chitin/chitosan based bio plastic applications (WP5) but also safety, environmental, social and economic sustainability assessments (WP6), a set of activities which are all highly relevant for future market exploitation (WP7).

A market-oriented approach will drive RECOVER R&D activities, capitalizing on participating bio-insecticulture (NUTRINSECT), food processors (MANOR), agriplastics manufacturers (NATURPLAS and the largest final users ASAJA & SAV), fertilizer producer (ENTO), novel technologies and monitoring solutions (IRIS) complemented by the AB, which includes FCC, SOTRAFA, REPSOL, ALGAE-ENERGY as large industries, BioCampo, CinG, METGEN and Carbiolice as SMEs and Remedia, CLT, PACKCLUSTER and ISWA as industry associations. In addition, Task 7.4.4 ensures that sufficient knowledge is collated to steer the research towards delivering a solution with good market potential. Our intention is to realize all the exploitation pathways for the results:

a) **Commercial exploitation**: a key exploitation route will be the industrial use of the project knowledge for the creation of new biotech solutions for plastics degradation, together with innovative bio-based and biodegradable materials products, within and beyond the validated applications (biofertilizers, bio agricultural films, food packaging), new processes and advisory services to deliver high competitiveness and growth impacts.

b) **Education and training**: the use of the new knowledge generated for upskilling and for enriching curricula in European education will have a high impact for raising the European knowledge base to prepare agricultural, food, packaging, fertilizers and other industry sectors to transition to the bio-based economy;

c) **Further research**: planning future spin-out research and development projects especially in areas that will contribute to the transferability of the project results to new applications (e.g. application of enzymes and microorganisms specifically inoculated insects (vectors) as means of biodegradation improvement and bioremediation of soils with different types of contaminants without generating problems of pests or diseases, etc);

d) **Contribution to standards**: while making available new end of life options for standard plastics, the project results could on the long term contribute towards a new and/or revised regulations and standards in the field of agri-food plastic waste and soil biodegradability

Detailed business plans will be developed in Task 7.4, Subtask 7.4.1 *Business and exploitation planning*, and annexed to the PEDR, in order to provide the details on how the project's results will be used in commercial exploitation activities, for training purposes, further research as well as standardization activities. Further measures will be planned in the framework of this task to increase the likelihood of continuity of the established ecosystem and value chains, as well as market uptake of RECOVER KERs for European industrial and primary sector sustainability, competitiveness, growth, and jobs. During the proposal stage, each partner has communicated clear exploitation expectations (Table 2.4), as well as expected growth derived from the proposal in line with the share of the respective previously discussed overall growth of the sector he represents and that he could tap in. This will be planned in greater detail during WP7 to maximize the innovation impact but while growing large companies by few percent already has a significant impact, it is especially massive for smallest biotech companies like ASA and NUTRINSECT.

Table 2.4. Summary of exploitation intentions of the RECOVER partners

No	Short Name	Exploitation intentions of the RECOVER partners	Turnover increase	New Jobs
1	UAL	Development of microbial consortia and insects for licensing. Use of the knowledge generated in further projects. Potential patent on procedure for plastics removal by microbial consortia	NA, non profit	
2	UNIPI	Exploit chitin/chitosan-based materials for active packaging, agriculture and health care products, exploit logistic system for plastic collection (licensing)	NA, non profit	
3	ASAJA	Exploit valorisation procedures to introduce as requirements in CAP (Common Agricultural Policy). Further development of procedures for bonus to the sustainable farmers who include Agri-plastics waste collect, pre-treatment and valorisation. Possible patent licensing specific plastic collect and pre-treatment (mechanical and biodegradation) machinery,	NA, non profit	
4	BRUNEL	Enzymes developed will be made available to project partners, and subsequently other commercial users under collaborative and licensing agreements, as appropriate for protection of IP and further enzyme development. IP will be developed, protected, and through Brunel's commercialization team, and relevant spin-outs (including Syngenious)	NA, non profit	
5	UMH	Development of procedures for fortification of earthworms with microbial/enzyme consortiums for licensing. Possible patent licensing specific plastic biodegrading systems including earthworms	NA, non profit	
6	ASU	Chitin extraction and Chitosan modification for packaging applications used in services and projects. Possible patent licensing on extraction process and modified materials.	NA, non profit	
7	NATURPLAS	Commercialization of new active mulching films.	7%	2
8	ASA	Commercialization of enzymes tailored for AWP degradation	25%	5
9	NUTRINSECT	Commercialization of insects tailored for AWP degradation	25%	5
10	IRIS	Further development (customization service) of the biodegradation predictive models and licensing of the derived app. Commercialisation of monitoring systems or specific chemiometrics package to be installed in existing waste sorting plants. Use of the knowledge generated in further projects.	5%	3
11	OWS (Terminated)	Increased knowledge on biodegradation of materials. Possible application of the process in the DRANCO and SORDISEP system developed and patented by OWS for treatment of mixed organic municipal waste.	6%	2-3
12	IDELUX	Exploit the technologies developed to improve the compost quality and reduce plastic environmental contamination.	NA, non profit	
13	FEMTO	Exploit chitin/chitosan-based materials for active packaging and agricultural products, possibility to promote its product as degradable by RECOVER approach	10%	3
14	ENTO	Commercialization of new chitin/chitosan fertilizers.	25%	5
15	MANOR	Commercialization of chicken products in new packaging and edible coatings.	>1%	2
16	ENCO	Use of the knowledge in bio-based plastics market in further related projects as well as in consultancy services.	10%	2-3
17	SAV	Use of the knowledge in fertilizers and improved mulching films validation to improve their agricultural practices. Use of RECOVER approach as a new source of revenues for the valorisation of EoL plastics and remediation of pollution.	5%	3
18	OWS RF	Increased knowledge on biodegradation of materials. Possible application of the process in the DRANCO and SORDISEP system developed and patented by OWS for treatment of mixed organic municipal waste	6%	2-3

The following table 2.5 gives a summary description of a selection of the expected RECOVER KERs:

Table 2.5. RECOVER Key Exploitable Results (Knowledge, products or technologies)

WP	KER	Owner(s)	Envisaged IPR Tool	Type of exploitation	Target end-users	Timetable
1	Monitoring system for waste materials	IRIS	Confidential know-how for hardware calibration + design, software secured by a code	Sales of turnkey solutions	Plastic waste managers	2 years after the project
2	Microbial consortia for plastic degradation	UAL	Patent or trade secret	Licensing	Organic waste treatment facilities and entities involved in bioremediation	2 year after the project
	Protocol for producing insects with plastic degrading capabilities	UAL	Patent or trade secret	Licensing	Insect producers	2 year after the project
	Novel degradative enzymes and design-knowledge	BRUNEL	Patent or trade secret	Licensing and spin-out	Farmers, waste management companies, plastics manufacturers, circular bio-based industries	2 year after the project
3	Protocol for producing earthworm with plastic degrading capabilities	UMH	Patent or trade secret	Licensing	Earthworm producers	2 year after the project
4	Predictive models and derived app for biodegradation optimisation, as well as biodegradation monitoring system	IRIS	Confidential algorithms that will be used in software development secured by code	Licensing, Sales of turnkey solutions	Farmers and plastics waste management companies	1-2 years once large Recover facilities are established for integration of the predictive model in a software product
	Biological process for treatment of waste polluted with plastics	OWS RF	Patent or trade secret	Licensing	Organic waste treatment facilities	2 years after the project
5	Chitin extraction and Chitosan modification for packaging applications	ASU	Patent or trade secret	Licensing	Packaging producers	1-3 year after the end of the project
	Active plastic based on chitin and or chitosan	UNIPI FEMTO	Patent or trade secret	FEMTO for its own products, possible licensing	Packaging and health care products	2 years after the project
6	Design and assessment services for the AWP processing full scale implementation	OWS RF ASAJA	knowhow	Provision of services to potential users	Farmers, MSW managers	Start supporting results roll out right after the project (possibly also via follow up funding rounds for early adopters taking the risks)

In order to demonstrate the affordability of the **investment for adopting the RECOVER ex situ biodegradation solutions by both farmers/agri-cooperatives and MSW management organizations,** the CAPEX/OPEX have been estimated, Business Case 1 (Fig. 2.6a) and Business Case 2 (Fig.2.6b), respectively.

For Business Case 1 (farmers/agri-cooperatives), an initial investment of €200.000 is required in Y0. This includes the physical assets (Installation costs; Pre-fabricated unit for the biodegradation of AWPs; Collection tools; Mechanical pre-treatment tools) to be acquired for the RECOVER set-up. The OPEX that are worth €50.000 include the expenses necessary to keep the RECOVER solution running (Annual amortization AWPs collection tools non-self-propelled; Annual amortization AWPs tools for mechanical pre-treatment; Initial insects population and yearly repositions; Microbial consortia; Novel enzymes; Logistics service; Workforce costs; Licenses, sanitary, gas, energy and water consumptions; Other indirect costs; Annual amortization - av. Type 10% - dependent on national law) over pre-fabricated unit for the biodegradation of AWPs). In addition, we have considered 10% of the CAPEX for the equipment maintenance.

In view of the expected revenues that will derive from the commercialisation of the high value chitin-chitosan-based formulations from the conversion of AWP in our target markets (biofertilizers, agricultural plastic products and food packaging), the Break Event Point (BEP - point at which total cost and total revenue are equal) is expected to be reached in Y3 as the profit starts to be positive (€50.770,00). Revenues have been estimated according to the added value products prices (€/kg) estimated in Table 2.1.
The overall benefits of RECOVER along the value chain will be much higher than the figures outlined here.

	Y0	Y1	Y2	Y3	Y4
CAPEX (€)	200.000,00	20.000,00	20.000,00	20.000,00	20.000,00
OPEX (€)		50.500,00	55.550,00	56.055,00	56.105,50
Total cost (€)	200.000,00	70.500,00	75.550,00	76.055,00	76.105,50
Progressive cost (€)	200.000,00	270.500,00	346.050,00	422.105,00	498.210,50
Prod FERT (ton/year)		30,00	31,50	33,08	34,73
Income FERTILIZERS(€/year)		60.000,00	63.000,00	66.150,00	69.457,50
Prod CHITIN (ton/year)		15,00	15,75	16,54	17,36
Income CHETIN (€/year)		90.000,00	94.500,00	99.225,00	104.186,25
Total revenue (€)		150.000,00	157.500,00	165.375,00	173.643,75
Progeressive revenue (€)		150.000,00	307.500,00	472.875,00	646.518,75
PROFIT (€)	-200.000,00	-120.500,00	-38.550,00	50.770,00	148.308,25

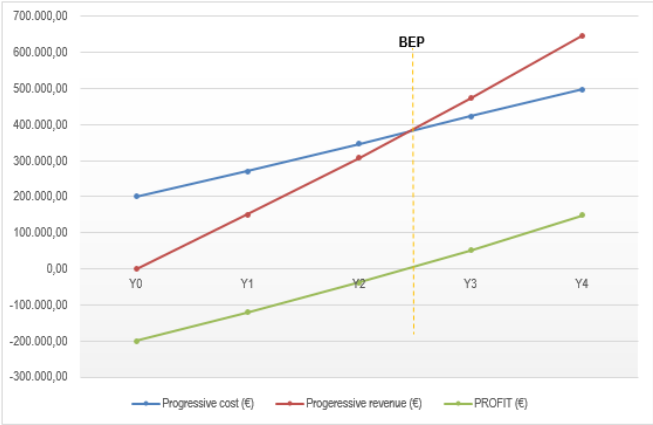


Figure 2.6a: Business Case 1 - primary producers - Investment breakdown to adopt RECOVER full scale AWP valorization process

For Business Case 2 (MSW management organizations), an initial investment of €1.700.000 is required in Y0. This includes the physical assets (as listed above but sized for a much greater treatment yield) for the RECOVER process set-up. The OPEX necessary to keep the RECOVER solution running are worth 1.500.000 (+10% of equipment maintenance). According to the expected revenues that will derive from the sales of chitin-chitosan-based formulations from the conversion of AWP in our target markets, the BEP is expected to be reached in Y3 as the profit starts to be positive (€15.500.000). Revenues have been estimated according to the price (€/kg) estimated in Table 2.1.
In both scenarios (Business Case 1 and Business Case 2) presented does not include the extra investments needed to reach TRL9, as it is part of post-project measures.

	Y0	Y1	Y2	Y3	Y4
CAPEX	1.700.000,00	85.000,00	85.000,00	85.000,00	85.000,00
OPEX		1.500.000,00	1.650.000,00	1.815.000,00	1.996.500,00
Total cost	1.700.000,00	1.585.000,00	1.735.000,00	1.900.000,00	2.081.500,00
TOTAL COST	1.700.000,00	3.285.000,00	5.020.000,00	6.920.000,00	9.001.500,00
Prod FERT (ton/year)	0,00	500,00	525,00	551,25	578,81
Income FERTT (€/year)	0,00	1.000.000,00	1.050.000,00	1.102.500,00	1.157.625,00
Prod CHITIN (ton/year)	0,00	200,00	210,00	220,50	231,53
Income CHETIN (€/year)		1.200.000,00	1.260.000,00	1.323.000,00	1.389.150,00
Total revenue (€)	0,00	2.200.000,00	2.310.000,00	2.425.500,00	2.546.775,00
Progeressive revenue (€)	0,00	2.200.000,00	4.510.000,00	6.935.500,00	9.482.275,00
PROFIT (€)	-1.700.000,00	-1.085.000,00	-510.000,00	15.500,00	480.775,00

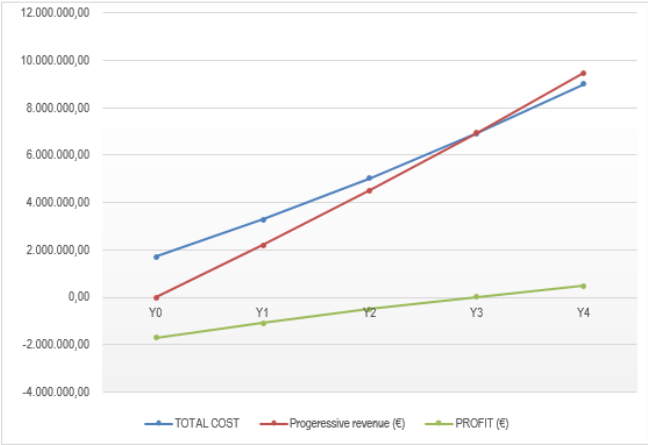


Figure 2.6b: Business Case 2- MSW management organizations - Investment breakdown to adopt RECOVER full scale AWP valorisation process

In line with the expected market penetration (0.1%) previously discussed in terms of the impact in the call linked jobs creation and using the expected revenues from table 2.6 a and b for each RECOVER process uptake, we expect our 220 agri-cooperatives and 400 MSW treatment organisations early adopters in Europe to generate annually close to € 724million turnover[150], €38.4 million net profit[151], and ~3.460 new jobs already by 2030 as previously discussed. The potential for further deployment in the

150 It derives from the expected revenues from agro-cooperatives (selling price is €200.000 paid by 220 agro-cooperatives and from MSW landfills (selling price is €1.700.000 paid by 400 MSW organizations).
151 Calculated as 10% of the expected revenues from agro-cooperatives (totalling €44.000.000, i.e. €200.000 x 220) + 5% of the expected revenues from MSW (€1.700.000 x 400)

broad agriculture and MSW sectors globally goes well beyond this.

In terms of the measures that will be in place in order to effectively plan for the exploitation of the results of the project, the robust industry involvement in the consortium (SMEs and Large companies), backed by strong knowledge management, knowledge transfer and market exploitation plans, and their key role in the project activities will boost an industry-driven approach to the project respecting market, end-users (farmers/MSW organisations) and regulatory requirements. This will ensure that RECOVER is successful in transitioning knowledge towards the market overcoming potential barriers that may hinder the swift uptake of the results of RECOVER research in the post-project phase.

Effective knowledge transfer, under *Subtask 7.4.4,* from technical partners to SMEs (internally) and end-users (externally) will also be fundamental for paving the way towards commercialization, guaranteeing that sufficient knowledge is assimilated and competencies to exploit results are acquired. To this end, the involvement of RTDs of international renown in the fields of polymers (UNIPI), microorganisms (UAL), *in silico* modelling (BRUNEL), vermicompost and worm monitoring (UMH), Chitosan extraction and use (ASU) and who have a philosophy of working closely with and in transferring research results to industry, as well as providing training and educational programmes, are value assets, which when combined with industry drive will ensure we bridge the innovation gap that frequently prevents industry adoption the results and R&D outputs. The non-industrial partners will exploit the results of RECOVER project to enrich their education offer with the novel concepts and system design principles that will be identified and developed during the project. Some of the topics tackled in the project shall become part of Master courses, as well as research topics for Master & PhD theses, offering students a taste of research problems that have practical impact and interest, and the opportunity to perform research in a high-quality international environment. Results from this project will be potentially reflected in the courses offered to students. Significantly the inclusion of a company with experience in market penetration (ENCO) is an important added value for the consortium and will contribute to ensuring a systematic and methodological approach to planning for the important post-project stage.

RECOVER consortium includes key partners, which have directly links and communications with end-users across Europe (COPA as AB) and Spain (ASAJA) as well as further multipliers in the AB (ISWA, PLACKCLUSTER…). Importantly, we will follow a cautious and transparent communication policy both in terms of sourcing and end of life, backed up by consequential LCA results, which will contribute to raising industry confidence that will allow us to deliver a positive breakthrough in the context of environmental-friendly plasticulture. This will be complemented by the lobbying power of COPA through ASAJA as well as of APE on policy makers towards the introduction of requirements in CAP policies for awarding farmers for AWP collecting and valorising. Sound management of the knowledge generated during the project, active planning in relation to post-project exploitation activities, further research work for escalating along the TRLs, as well as scouting for possible viable financing routes to commercialization that will allow us to quickly scale up to complete the bridging to the Innovation Gap and the subsequent navigation of the 'Valley of Death', to swiftly overcome the critical stage between pre-competitive results and the market. All this will be complemented by a solid communication strategy to distribute non-confidential results to relevant audiences as well as to market actors, multipliers and policy makers.

How the participants will manage the research data generated and/or collected during the project

Although the exploitation interests will be favoured by default when considering the project results, the RECOVER project will participate in the Pilot Open Access Research Data scheme for a selected amount of research data generated by the project. As such a data management plan has been included among WP8's deliverables (D8.5). It guarantees that research data will be made findable, accessible, interoperable and re-usable (FAIR) and also instrumental for the IP and dissemination action. Non-confidential results will be widely disseminated in Open access media as further discussed (publications, selected biodegradation, LCA and characterization results will be considered among others to provide the scientific community with understanding of e.g. the operational conditions to enhance bioremediation process, the biodegradation modelling after an agreed embargo time allowing excluding the commercially sensitive details). All publicly available data will be protected by copyright statements and some with commercial values could be available under a fee. Data curation and preservation will be ensured by an intranet with a specific platform to be developed as part of the project website, accessible only to its partners. In it, all the necessary information will be uploaded in the appropriate formats for its optimal use. In this context, this information will be kept for a maximum period of two years after the end of the project. Moreover, all data exchange transactions (including external surveys for the social acceptance study) will be made in accordance with the EU General Data Protection Regulation (GDPR). In terms of data curation and preservation management, ENCO is responsible for the managing of the intranet, and therefore the information contained therein.

Figure 2.7: RECOVER Knowledge Management Strategy

Strategy for knowledge management and protection

The Knowledge Management Strategy (Fig. 2.7) which will be followed during the RECOVER lifecycle is heavily interconnected with the exploitation and dissemination activities to pave the way forward for post-project exploitation. An integrated approach will be followed for the identification, capture, retrieval, distribution, sharing, use and reuse of RECOVER information and knowledge assets. This will be essential to drive the value and benefits of these information and knowledge assets, and contribute to maximise the innovation impact of the action. Knowledge Management sessions will be taken place regularly during the project meetings. These sessions will be chaired by the IPR & Exploitation Manager (Dr. Marco de la Feld, ENCO) attended by all project beneficiaries and will facilitate the disclosure of the IP and the novelty of the project results. The aim of these sessions is to review: i) Expected results per Work Package (WP). ii) Evaluation of the IP (technology watch/prior art review). iii) Evaluation of protection options (IPR tools evaluation). iv) Third party components included in the project activities including tools and datasets. v) Confidentiality of project results. vi) Foreseen publications per partner. vii) Exploitation intentions per partner.

Background IPR and access rights

Background- information and knowledge (including inventions, databases, etc.) held by the partners prior to their accession to the EC grant agreement, as well as any intellectual property rights which are needed for carrying out the project or for using foreground held by the consortium, or indeed by third parties- is defined in the proposal preparation phase (pre-Grant Agreement). The current status of the Background being brought to the project by the consortium is listed in Table 2.6. Access to Background that is, or will be found to be, necessary for the implementation of the project will be granted royalty free and background needed to use the results of the project will be provided either royalty free or under fair and reasonable conditions.

Table 2.6. Background per partner

No	Short Name	Background
1	UAL	Know-how in microorganism selection and testing for bioremediation of recalcitrant molecules and xenobiotics and microbiology of composting. Know-how on insect studies at lab scale
2	UNIPI	Knowledge on polymer characterization, know how in polymer processing and processing of chitosan and chitin and their application in packaging and health care.
3	ASAJA	Knowledge in AWs mixtures at farm and MSW level
4	BRUNEL.	Knowledge in comparative genomics and outputs applicable to novel enzyme selection and synthesis
5	UMH	Know-how in earthworm uses in bioremediation (degradation of polyphenols, pesticides and other pollutants); Know-how in composting, vermicomposting and biochar
6	ASU	Know-how in extraction and formulation (including functionalization) and processing of biopolymers for packaging applications.
7	NATURPLAS	Knowledge in market and requirements from agriculture films
8	ASA	Know-how in enzymes, enzyme technology, fermentation, down-stream processing
9	NUTRINSECT	Know-how in insects and industrial production of insect-based new materials
10	IRIS	Know-how in monitoring systems for waste composition detection especially including NIR, HSI -Knowledge on data mining for predictive modeling.
11	OWS (Terminated)	Knowledge on biodegradation tests in all kind of environments according to international standards and on the analysis of organic recyclability such as compostability (e.g. the complete EN 13432 testing program). Know-how on improving biodegradation test procedures. Know-how on LCA.
12	IDELUX	Industrial experiences of biowaste treatment, knowledge and experience of the plastic waste stream and know-how in composting and plastic contamination
13	FEMTO	Know how in polymer processing and recycles
14	ENTO	Know-how in formulation of fertilizers from chitosan. Know-how in BSF (black solider flies) valorised bio-fertilizers from wastes
15	MANOR	Know-how in packaging and edible coatings.
16	ENCO	Knowledge in communication, business & market analysis. Knowledge in innovation, exploitation and IPR management
17	SAV	knowledge in fertilizers and mulching films validation for agricultural and gardening practices.
18	OWS RF	Knowledge on biodegradation tests in all kind of environments according to international standards and on the analysis of organic recyclability such as compostability (e.g. the complete EN 13432 testing program). Know-how on improving biodegradation test procedures. Know-how on LCA.

Results definition, protection, ownership and access rights

Selected results from the project were previously listed in Table 2.5 along with the expected scenarios of results ownership. This table will be fully completed for all project results, and it will be periodically revised and updated during the project in line with developments. The project results will be continually assessed in terms of novelty, patentability and protection through T7.3, as part of an effective knowledge management strategy. The choice of the most suitable form of IP protection, as well as the duration and geographical coverage depends on the results at stake, but also the business plans for their exploitation and legitimate interests of consortium partners. By default, results will be owned by the partner generating them. To avoid doubt or confusion over ownership, a protocol of procedures will be defined, which will be followed by each partner to clearly show the development of the generation of knowledge or results. In case where a result is jointly generated (i.e. where the separate parts of some results are attributed to different participants), it will be jointly owned. Consequently, Joint Ownership Agreements will regulate the allocation and the terms of exercising this joint ownership(s). However, an overriding principle will be that the participating industry partners are always provided with the access that they need to the results to exploit them in keeping with the needs and strategy of their business. As such, access rights to results for implementing the project will be granted on a royalty-free basis. Access rights for use purposes will be granted either royalty-free, or on fair and reasonable conditions (these conditions will be established in Exploitation Agreements that will be negotiated between the requester and granter if the access rights). A first draft of the Exploitation Agreement will be available by M18 (D7.3) and will be continually revised and edited throughout the project, and a final version will be signed by all partners at M48 (annexed to D7.5).

Open access to peer-reviewed scientific publications

RECOVER partners will provide peers and end-users with free of charge on-line access to the project peer-reviewed scientific research articles. Partners can follow two routes towards open access **(1) by self-archiving their journal articles in an open access repository, also known as 'green' open access, or (2) by publishing in an open access journal, known as 'gold' open access.** Whereas fees to cover publishing in Gold open access journals have been budgeted, RECOVER will privilege green open access, whereby the partners will publish in peer-reviewed scientific publications and then self-archive a version in repositories (using only those indexed in specialized search engines like Scopus). The project website will complementarily house the 'RECOVER Green Channel' for open access to the peer-reviewed publications of the project. In relation to expected publications from the project, these include and are not limited to: results from demonstration of the pre-compatibility of micro and macro-organisms consortia for plastic biodegradation; results from the upscale of ex and in situ biodegradation process; results from value chains analysis; LCA, s-LCA and LCC; results of the testing of the different developed added-value downstream products (biofertilizers, chitin and chitosan derived bioplastics and coated materials), etc.

b) Communication activities

Communication activities are of the outmost importance for effectively promoting our action and outputs towards the stakeholders that we want to reach though this BBI action. To successfully connect, cooperate with and acquire positive support of our public a strategic plan will be implemented in Task 7.5, on top of the preliminary outline drawn up in Table 2.7. The overall communication campaign, which covers the whole project duration, will be led by the Communication Manager, Mr. Juan Almansa (ASAJA), who already holds an established extensive network and communication capacity to ensure that benefits and impacts of the project are genuinely conveyed to key stakeholders and general public. Our aim is to i) generate understanding about what RECOVER is seeking to achieve, ii) stimulate interest and support, and iii) motivate the collaboration of actors along the value chain as well as of general public in order to boost the transition to the bio-based economy in EU.
RECOVER communication campaign leverages one-way and multilateral strategic channels, following these criteria: Why – Outcomes expected; What will be communicated – the message; To whom – the audience; How – the method; When – the timing.
Following a preliminary stakeholder analysis (that will be exhaustively developed in Task 7.2), the target audiences of RECOVER communication activities comprise Agricultural Cooperatives; Farmers; Insecticulturers; Plastic manufacturers and converters; Packaging industry; Food manufacturing industry and retailers; Fertilizers companies; Soil bioremediation and waste management companies, Consumers & Consumer organizations; Public authorities and Policy makers; Standardization & Safety Agencies/Organizations; Scientific communities; Particular efforts will be paid to engage with and raise awareness of the general public, also to gain social acceptance. It is important to address the public concerns by informing the public on our strong dependence on conventional plastics which shall be adequately handled at EoL through innovative biotech solutions and on the advantages to choose innovative bio-based materials adjusted to the applications. The awareness actions carried in RECOVER will aim to generate acceptance for our different value-added products but also towards insecticulture which is a growing arena and here fundamental to maximizing the enzyme and microorganisms efficiency for plastic degradation through optimal communication and transparency using appropriate language. A set of key messages will be developed specifically to popularize complex subject and to bring the issue closer to people's lives by making it more understandable. The campaign will be rolled out through various channels and tools (table 2.7), with the following approach: 1. 'Push'

communication (graphic identity, newsletters, press releases, promotional materials, gadgets, etc) and 2. 'Pull' communication and interactive (website, social media, liaison with other initiatives, joint communication, etc). The fact that our demonstrators are in part focused on consumers applications like packaging and in the broad part on plasticulture will allow maximising the awareness of the impact of bioeconomy, and indeed BBI programme, on their everyday life. In RECOVER, collaboration with the European Climate Adaptation Platform (Climate-ADAPT)[152] and agricultural European Innovation Partnership EIP-AGRI[153] will be particularly pursued to magnify the replicability of the project results, by sharing the project experience and lessons learned with stakeholders while simultaneously learning how these actors are managing and reacting to similar problems focused primarily on the problem of AWPs. In addition, to provide added value to the RECOVER end-users, EIP Multiactor Practice Abstracts will be prepared (in all partner languages) - 1.000 to 1.500 characters) highlighting main practical recommendation (how can practitioner make use of the RECOVER results).

Table 2.7: RECOVER Communication Toolbox and Approach

Approach	Measure	Timing	Impact	Key Metrics
Push Effect	Logo & Project Identity	M1	Recognizable project identity that will help increase awareness about the project	At least 1 logo, 2 templates, 1 set of imagery
	E-Newsletter	6-monthly	Create community sense & Give visibility to events and activities related to the project.	At least 150 enrolments per year
	Press Releases & Media presence (TV, Radio)	6 per year	Achieve maximum awareness of the project & share project vision	At least 5 press-releases, 10 articles and 5 interviews in media (Radio/TV)
	Digital/Printed Material	M1-M12-M24-M36-48	Create awareness of the project. Spread project results.	At least 3 flyers designed & 1500 printed; 3 posters; 3 infographics
	Videoclip	M30	Showcase benefits and results	At least 1000 visualizations
Pull Effect	Project Website	M1, regularly updated	Create awareness, disseminate information about the project.	At least 5k unique visitors
	Social Media	M2, content online	Reach & engage different target audiences	At least 1000 fans
	Interaction with ongoing initiatives/projects	Linking projects, platforms and database	Synergies with cross-cutting initiatives and cross-fertilization	At least 3 officially MoU signed for cross-fertilization
	Participation to events & conferences	At least quarterly from M3	Increase awareness of the project and its results	At least 2 events per partners
	Final Conference	M47	Engage public & stakeholders to amplify outreach of RECOVER	At least 100 attendees

3. Implementation

3.1 Work plan — Work packages, deliverables and milestones

Brief presentation of the overall structure of the work plan

A competitive work plan has been devised in order to deliver on the objectives of the RECOVER project providing tailored end of life solutions for a number of AWP using innovative biotech approaches. This will preserve our environment and resources but also meet the society and regulatory demands. The project objectives have been used as the starting point for planning the work that needs to be implemented within the project (Section 1.1). All the actions that are required to meet the objectives of this Research and Innovation project have been collectively analysed by the interdisciplinary consortium. It was agreed how best to distribute the workload among the various experts in keeping with their relevant fields of knowledge, capacity, infrastructure, resources and in the context of bioplastics supply, packaging textile industry value chains, as well as waste management companies. This bottom-up planning process has greatly facilitated the task of arriving at a clear, comprehensive and realistic work plan (see Gantt Chart and Pert Diagram below), whereby all partners their ownership.

To fulfilling the project objectives, our work plan is split into 8 WPs with clear tasks that will enable the project's objectives to be fulfilled through a coherent structure that will also facilitate project. To ensure the clear focus towards the target application of this Research and Innovation Action building upon past successful research, most WPs are led by industrial partners. WP1 deals with industry-led specification in terms of AWP sourcing to match the market needs. WP2 deals with the lab scale development of the new biotech tools to degrade AWP and WP3 with the upscaling of their production and their combination. WP4 comprises field trials and large scale composting trials to validate the new process in an industrially preliminary scale.

152 https://climate-adapt.eea.europa.eu
153 https://ec.europa.eu/eip/agriculture/